OpenBoatLink (OBL) – Open Radio Protocol Specification, Draft v0.6

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Motivation

Advanced electronics can make regattas and training more attractive for sailors and spectators: live tracking and replay, electronic starts with on-board countdown and OCS detection, umpiring support, telemetry for coaching, and automated safety check-in. Most events do not use them yet. Where they are used, each application brings its own proprietary radio and platform, the systems cannot share data, and cost and effort limit them to a few top events.

OpenBoatLink (OBL) is one open, royalty-free radio protocol for all these applications. Devices from different manufacturers interoperate without prior coordination. OBL specifies the physical layer, medium access and a compact message set. It builds on techniques proven in aviation, mining and construction (see System overview), adapted to the ranges, speeds and fleet sizes of sailing. Three performance tiers let simple, affordable devices serve club and national racing, while Olympic and world championship events get centimetre-level positions and accurate headings, precise enough to decide starts, finishes and mark roundings. The core of the specification is also published as an open-source C library. Its modules that decide compatibility and security are used unchanged in every product, which keeps devices interoperable and simplifies testing; the other modules are recommended.

World Sailing recognises such systems as Augmented Race Management Systems (ARMS) under Policy D12. OBL covers the ARMS functions with one radio link and adds an open protocol, so that recognised devices of different vendors can be used together at one event. Appendix E maps the Policy D12 requirements to this specification.

Use cases

- Tracking: live positions of the fleet for race committee, media and spectators; replay after the race.

- Race management, as Augmented Race Management System (ARMS): start sequence with GNSS-synchronized timing; start line position and distance to the line on board; OCS detection and notification of the boats concerned; course changes and other signals; finish detection with automatic scoring.
- Umpiring: positions, headings and turn rates in real time to support on-water umpire decisions; penalty and exoneration signals to the boats concerned.
- AI umpiring: automated rule checks (overlap, mark room, right of way) based on high-rate tracking data.
- Safety check-in/check-out: automated presence control before, during and after racing; distress signalling (capsize, man overboard); geofencing.
- Coaching and telemetry: wind, heel and trim, rig loads and other sensor data, live on the water and recorded for debriefing and performance analysis.
- Race mark monitoring and management: position, drift, wind data and battery state of anchored and robotic marks.
- RTK correction broadcast: GNSS correction data for centimetre-level positioning of competitor boats and marks, improving all ARMS position data: start and finish lines, OCS identification and mark roundings.

Requirements

- Scale: full performance for up to 150 boats within radio range; graceful degradation up to 600 boats; no collapse beyond.
- Direct reception, low latency: boats, umpires, coaches and race committee receive each other directly, within a second and without shore or internet infrastructure.
- Licence-free operation: sub-GHz ISM/SRD bands (868 MHz in Europe, 915/921 MHz in the Americas and Oceania) within regional power and access rules.
- Scalable performance: three performance tiers, from simple devices for club racing to RTK devices for Olympic events, all on the same protocol (see OBL performance tiers).
- Cost and power: complete electronics of a standalone device USD 20-30 in bill of materials; less than USD 10 to add OBL to a device that already has GNSS. Several race days on a small battery or solar power.
- Zero configuration: boat devices are not configured for a regatta or a course. The only exception is a semi-permanent boat class setting.
- Common hardware: every component is a common part, available from at least two manufacturers.

- Robustness: reliable over-water links across a course of several kilometres, with forward error correction.
- Authenticity: every receiver can verify sender, integrity and freshness of race management, correction and tracking data, without shared secrets or shore infrastructure. An event can encrypt boat data (see Encryption).
- Openness: public, royalty-free specification. Anyone may build receivers, and transmitters with an assigned address block.
- Reference library: the modules of the official C library that decide compatibility and security are mandatory and used unchanged in every product; the remaining modules are recommended (see Conformance disclosure).

OBL performance tiers

OBL defines three performance tiers for devices, from tier 1 (highest) to tier 3 (lowest), like the grading of events. Tier 3 allows simple, affordable devices for club racing and training; tiers 2 and 1 serve championships and Olympic events. All tiers use the same protocol and share the same channels. A tier describes how well a device measures its position and how its data may be used, not how it communicates.

Tier	Suggested events	GNSS, accuracy and fix rate	Heading	Authentication and encryption	ARMS use
1	Olympic Games, high-speed events	Centimetre level: RTK with the venue correction stream; fix rate at least 10 Hz	Required, within 2° (draft)	Required	Category 2 for all decisions
2	World and continental championships, international regattas, national championships	About 1 m: multi-constellation, single- or multi-band; accuracy reported with every fix; fix rate at least 10 Hz	Required, within 5° (draft)	Required	Category 2 for timing and OCS; result over, clear or too close to call
3	Club racing, training	Any GNSS; fix rate at least 1 Hz	Optional	None; no key store	Tracking; Category 1

The suggested events are guidance only; the notice of race or the sailing instructions state the minimum tier. A device meets a tier if it meets all requirements of that tier. It declares its tier in its information message (part 0), so that the race committee can check

the fleet over the air.

Tier 1 and 2 devices shall measure the heading, normally with a magnetometer and IMU. Dual-antenna GNSS is an option where two antennas can be placed 20 cm or more apart, for example on yachts or on dinghies one at the stern and one in the display unit near the mast. OCS and zone calls place bow and stern from antenna position and heading; at low speed or in current, the course over ground does not show where the boat points. The heading limits in the table are provisional; they will be set with the manufacturers of heading units after trials on the water. They apply to the validity check described under Flags (hdg_valid). Each device reports its current accuracy in its tracking messages (pos_acc). The default deviation threshold of extrapolation-based reporting follows the tier: 0.2 m at tier 1, 1 m at tier 2 and 2 m at tier 3. The fix rate is the GNSS fix rate. Each device checks every fix against the prediction, so a higher fix rate detects manoeuvres sooner; tier 1 and 2 devices also record every fix on board (see On-board recording). The rate of transmitted reports follows from extrapolation-based reporting and the operating points (see Capacity and scaling).

Tier 3 devices need no key store. They do not authenticate or encrypt their own messages, but still verify authenticated infrastructure messages in software. Tier 1 and 2 devices authenticate their messages and encrypt them when the event requires it (see Authentication and Encryption). World Sailing ARMS Category 2 (rule-determining) depends on the decision. Timing decisions, such as the official start time, depend only on GNSS time and authenticated race committee messages. Position-based decisions, such as OCS identification, use the start report of the boat: it gives the margin to the line and the accuracy of the position, so the result is over, clear, or too close to call, and the race officer decides the last group (see Race management message). Tier 1 and 2 devices are suitable for both. Tier 3 devices do not authenticate and are suitable for Category 1 (decision support) only.

Each tier also requires a minimum conformance: the following subtopics shall have the status Library or Complete in the device's Implementation Conformance Statement (see Conformance disclosure).

Tier	Minimum conformance (ICS: Library or Complete)
1	All subtopics, including RTK
2	All subtopics, except RTK and optional telemetry subtypes

Tier	Minimum conformance (ICS: Library or Complete)
3	Shared-channel subtopics (Modulation and modes, Channel plan, Timing, Medium access control), tracking and information messages

At every tier, the mandatory library modules shall be used unchanged (see Conformance disclosure). A device shall declare a tier in its information message only if its published ICS supports that tier. Once test vectors and conformance tests are published, a tier counts only for devices that have passed them.

System overview

OBL builds on techniques proven in other fields: electronic conspicuity of aircraft in the 860 MHz band (EASA ADS-L 4 SRD-860) and the tracking protocol of the Open Glider Network (OGN); collision warning systems for mining and construction machinery (requirements: ISO 21815); and dead reckoning with report-by-exception from distributed interactive simulation (IEEE 1278.1). See References. For race management, OBL provides the functions of an Augmented Race Management System (ARMS), the World Sailing term for electronic start timing and OCS identification.

Components

- Boat node: GNSS receiver and sub-GHz transceiver on each boat, standalone or built into other equipment such as race computers or instruments. Sends tracking, information, telemetry and start report messages; receives race management and RTK messages.
- Infrastructure nodes: equipment of the organizing authority. Committee node (race management, optional RTK base node), mark nodes on anchored or robotic marks (tracking, telemetry, weather), umpire, coach and rescue boats (receivers, optional relay nodes).
- Relay nodes: infrastructure nodes that retransmit selected packets once to extend coverage. Boat nodes do not relay.
- Gateway nodes: receivers that forward radio traffic to the internet and inject shore traffic into infrastructure slots (see Gateways).
- Backend: aggregation, live tracking, storage and replay. Out of scope of this specification.

Principle of operation

All devices are synchronized to UTC by GNSS. Airtime is divided into one-second cycles of four 250 ms frames. Two high bandwidth (HB) frames carry most of the traffic. Two long range (LR) frames give range to boats with low antennas and infrastructure. HB and LR frames use separate channels, chosen per region to match its access rules. Each frame has fixed slots (1 ms in HB, 4.25 ms in LR). Every packet is 232 bits long and fits one slot.

Devices choose free slots based on what they receive (listen-before-talk and an occupancy map) and keep a slot that works (sticky slots). The fleet converges to a low-collision pattern without registration, pairing or a master station.

A position takes 40 bits at 4.44 cm resolution (relative coordinate encoding). Speed and rate of turn use nonlinear exponential encoding. Each device predicts its own track from its last transmitted report and sends an extra report only when the actual position deviates from the prediction (extrapolation-based reporting). Receivers use the same prediction between reports. Under high load, boats on steady courses report less often and leave airtime to boats that manoeuvre. The chapter Capacity and scaling gives the budget.

Zero configuration

Boat devices are not configured for a regatta or a course. They take all settings from three sources:

- GNSS position: selects the regulatory region, and with it the channel plan and power limits.
- Channel load, measured in the occupancy map: sets reporting rate and deviation threshold.
- Event settings, broadcast by the organizer's infrastructure (race management message, subtype 5): event identifier, authentication and encryption mode, HB channel set, rate limits, relay request and course area geofences. Course areas do not overlap; a device selects its course area by position.

Without infrastructure in range, a device uses defaults. Only the organizer's equipment (committee boats, marks, relays, gateways) is configured. The one exception on boats is the boat class: a device may be set semi-permanently to the class of its boat. It sends the class in its information message and ignores race management messages for other classes.

Conventions

Requirement language

“Shall” marks a requirement, “should” a recommendation and “may” an option.

Nodes and devices

A node is a participant in the protocol, named by its role: a boat node, or an infrastructure node, which is a committee, mark, relay, gateway, RTK base or land-station node. A device is the physical product that implements a node; one device is one node. This specification says node where it describes behaviour on the air, and device where hardware, certificates, performance tiers or conformance are concerned.

Directions

All directions refer to true (geographic) north. Course over ground (COG) is the direction of movement over ground, from GNSS. Heading is the direction the bow points, from a magnetometer with IMU or from dual-antenna GNSS (see OBL performance tiers); the two differ with leeway, current and at low speed. Wind directions are the direction the wind comes from; true wind is ground-referenced, as opposed to apparent wind.

Library tags

The official OBL library, an open-source C library, implements the core of this specification in modules. Sections that a module implements carry a tag under their heading:

- Library: mandatory. Products shall use this module unchanged, from a released library version. It decides compatibility on the shared channel or the correctness of authentication and encryption.
- Library: recommended. Products may use this module or their own implementation that passes the conformance tests for the section.
- No tag: the section is not implemented by the library (hardware, policy or informative content).

The module list and the rules are given in Conformance disclosure.

Bit and byte order

- Hexadecimal numbers have the prefix 0x, binary numbers the prefix 0b.

- A byte has 8 bits. Bytes are numbered in transmission order; byte 0 is sent first.
- Within a byte, bit 0 is the least significant bit (weight 0x01) and bit 7 the most significant bit (weight 0x80).
- Multi-byte values and values that cross byte boundaries are little-endian: the least significant part is sent first.
- On air, each byte is sent most significant bit first. This is normally a transceiver setting.

Field positions in tables are bit offsets from the start of the block (header or payload).

Coordinate datum

WGS-84 is used throughout. Latitude and longitude are signed integers in units of 1E-7 degrees (raw format), north and east positive. One raw unit is about 1.11 cm at the equator. Messages carry no altitude.

Code fragments

C and Python fragments are informative. They show the algorithms but are not production code; implementations need error handling and defensive coding.

Limitations

- This document is a draft. All parts are subject to change
- OBL is a sports and safety-support system. It is not a navigation or collision-avoidance system under COLREG and does not replace AIS. It shall not be the only means to locate a person or vessel in distress.
- Latitudes of 79° and above are not supported (see Relative coordinate encoding).
- Payloads are sent in clear unless the event encrypts boat data (see Encryption). CRC and FEC protect against transmission errors only. Protection against forged, altered and replayed messages comes from authentication (see Authentication).

Open items, to be resolved before submission to World Sailing:

- Regulatory compliance of the HB mode in the EU and USA, and the EU power limit for wideband signals (Modulation and modes, Channel plan)
- Forward error correction: keep or drop (Forward error correction)
- Certificate format (Trust chain, Encryption)
- Test vectors and conformance tests (Conformance disclosure)
- Heading limits per tier (OBL performance tiers)

- Capacity simulation, including three fleets starting together (Capacity and scaling)
- Governance of the registry and the reference certifications (Governance)
- Range tests for devices with integrated antennas, mounted low over the water (skiffs, boards) (Antenna and installation)

Radio system

Packet structure

Library: mandatory (obl_frame)

All packets have the same length of 232 bits. There are no variable-length packets, which keeps the slot grid and the decoder simple.

Block	Length	Content
Preamble	16 bits	0b1010…, not whitened
Sync word	24 bits	0xB4 0x2B 0x59
Header	32 bits	Address, message type, urgency, relay budget (see Appendix A)
Payload	96 bits	Message-type specific (see Appendix A)
CRC	16 bits	CRC-16/CCITT over sync word, header and payload
FEC parity	48 bits	LDPC(192,144) parity over header, payload and CRC
Total	232 bits	773 µs on air at 300 kbit/s (HB)

Modulation and modes

Library: recommended (obl_hal, drivers for the reference transceivers)

The modulation is 2-GFSK with Gaussian filter BT = 0.5, in two modes. The high bandwidth (HB) mode carries most of the traffic. The long range (LR) mode trades airtime for sensitivity: dinghies carry devices with antennas of 9 cm or less, about 0.5 m above the water. Over water, the path loss between two such antennas grows with the fourth power of distance, so the HB mode cannot cover a race area from boat to boat.

Both modes use the same preamble, sync word, whitening, FEC and CRC. They differ only in these parameters:

Parameter	High bandwidth mode	Long range mode
Bit rate	300 kbit/s	62.5 kbit/s
Frequency deviation	0: −75 kHz, 1: +75 kHz (modulation index 0.5)	0: −15.625 kHz, 1: +15.625 kHz (modulation index 0.5)
Channel spacing / RX bandwidth	1 MHz / approx. 450 kHz	1 MHz / approx. 110 kHz
Packet airtime / slot duration	773 µs / 1 ms	3712 µs / 4.25 ms
Sensitivity	reference	approx. +7 dB gain
Max. radiated power	Europe: 25 mW ERP; USA, Australia, NZ: regional limits	Europe: 25 mW ERP, infrastructure up to 500 mW ERP; USA, Australia, NZ: regional limits

No Manchester coding is used. Whitening provides DC balance and clock recovery; the FEC provides error performance.

300 kbit/s and modulation index 0.5 (about 450 kHz occupied bandwidth) are the limits of common sub-GHz transceivers. Compliance with the minimum bandwidth rules of the EU (wideband data) and the US (digital modulation) is an open item of the regulatory review.

LR packets are sent only in frames F2 and F3; frames F0 and F1 always carry HB packets (see Timing). All devices shall switch their receiver configuration at frame boundaries and receive both modes.

Devices with antennas lower than about 1 m above the water shall send periodic tracking in LR slots and deviation reports in HB slots, so that the scarce LR slots carry only the periodic load; the next periodic LR message restores the picture for the whole course. Devices with elevated antennas use HB for all traffic. Mixed fleets share each cycle.

LR slots are reserved for traffic that must reach the whole course: tracking (msg_type 0), RTK correction (3), race management (4), safety (5), weather (6) and the authentication messages (7) of infrastructure nodes. Information (1), telemetry (2) and the

authentication messages of boat nodes shall be sent in HB frames only. A device may use both modes, for example tracking in LR and telemetry in HB.

The LR mode requires a carrier frequency accuracy of ±2 ppm or better. LR devices shall use a TCXO or discipline their frequency reference to GNSS.

Channel plan

Library: mandatory (obl_sched)

The channel plan is defined per regulatory region. Devices select the region from their GNSS position. HB frames and LR frames use separate channels, so that each traffic type can use the band whose rules suit it best. LR channels are common to all course areas on purpose: they carry safety and distress, race management, RTK corrections and the periodic tracking of low-antenna boats, which should reach every course area and every rescue or committee boat without listening on several channels. HB frames use one of up to four HB channel sets; set 0 is the default. The organizer assigns a set per course area in its infrastructure configuration and announces it in the event settings (hb_set), so that fleets starting at the same time on nearby course areas do not share the HB capacity. The plan is a draft and subject to regulatory review.

Region	HB channels (F0, F1)	LR channels (F2, F3)	Access rules
Europe	865-868 MHz; sets 0-2 draft 866.5, 865.9 and 865.3 MHz; F0 and F1 on the same channel	869.4-869.65 MHz; draft 869.4625 and 869.5875 MHz	HB: 1 % duty cycle, 25 mW ERP (for signals wider than 200 kHz possibly 10 mW; to be confirmed). LR: 10 % duty cycle, up to 500 mW ERP. The RTK base node uses the LR band (see RTK correction message).
USA / Canada	902-928 MHz, frequency hopping over at least 25 channels (draft: 50 channels, 500 kHz spacing); set n offsets the sequence by 12 n channels	902-928 MHz, frequency hopping over at least 50 channels (draft: 64 channels, 200 kHz spacing)	FCC §15.247 frequency hopping: at most 0.4 s per channel within 10 s (HB) or 20 s (LR); 20 dB bandwidth at most 500 kHz.

Region	HB channels (F0, F1)	LR channels (F2, F3)	Access rules
Australia / NZ	915-928 MHz, frequency hopping; sets as in the USA	915-928 MHz, frequency hopping	LIPD class licence; channel counts and power to be confirmed.

In Europe the HB channels avoid LoRaWAN (867.1-868.5 MHz), high-power RFID and the audio band 863-865 MHz; the LR band is shared with Meshtastic and LoRaWAN downlinks at 869.525 MHz.

With frequency hopping, each frame uses one channel of the region's hopping sequence. The channel follows from the Unix second and the frame number through a published pseudo-random sequence, so all devices of a channel set hop together without signalling, and each channel is used equally. Carrier sense is used in all regions (see Medium access control). Duty cycle limits are given in Regulatory duty cycle.

Timing

Library: mandatory (obl_sched); PPS capture: recommended (obl_hal)

All devices use their GNSS receiver as UTC time reference. Transmitting devices shall keep a time base accurate to ±5 µs. This requires capturing the GNSS time pulse (PPS) with a hardware timer; software timestamps are not sufficient.

Devices without a valid time base shall not transmit. Exception: a device in distress (urgency 3) that loses its time base, for example with the GNSS antenna under water after a capsize, may continue to send distress tracking messages with its last valid fix for up to 60 s on its free-running oscillator. Occasional collisions from the growing timing error are accepted.

A cycle is one second long, aligned to the UTC second, and has four frames:

Frame	Start	End	Channel	Mode
F0	0 ms	250 ms	HB channel of the set	High bandwidth
F1	250 ms	500 ms	HB channel of the set	High bandwidth
F2	500 ms	750 ms	LR channel 1	Long range
F3	750 ms	1000 ms	LR channel 2	Long range

Every frame starts with a retuning gap, in which the transceiver changes channel and mode. HB frames have 250 slots of 1 ms, numbered 0-249; slot 0 is the retuning gap and is never used. LR frames start with a 3.5 ms retuning gap, followed by 58 slots of 4.25 ms, numbered 0-57. HB slots 1-7 and LR slots 0-3 are reserved for infrastructure nodes: race management, relay and gateway nodes, and the start-line nodes around a start (see Gateways and Race management message). LR slots 4-11 are RTK slots: they are reserved for the RTK correction stream while the event settings announce an active stream (rtk = 1), and are contention slots otherwise. All other slots are contention slots.

Each slot is structured as follows. The transmitter starts at the packet offset and shall be off before the guard time begins.

Part of slot	HB slot	LR slot
Carrier sense (CCA)	0-40 µs	0-200 µs
Receive-to-transmit turnaround	40-140 µs	200-300 µs
PA ramp-up	140-155 µs	300-316 µs
Packet	155-928 µs	316-4028 µs
PA ramp-down	928-943 µs	4028-4044 µs
Guard time	943-1000 µs (57 µs)	4044-4250 µs (206 µs)
Slot duration	1000 µs	4250 µs

The guard time covers the timing error of sender and receiver (2 × 5 µs) and the propagation delay (3.3 µs/km, 33 µs at 10 km). Turnaround and ramp times are draft values, to be confirmed with reference transceivers. A transceiver that needs more time shall start carrier sense earlier, in the guard time of the preceding slot, and shall not transmit late.

Position, speed and heading in a packet refer to the time of the GNSS fix given in the time field, not to the time of transmission.

Medium access control

Library: mandatory (obl_sched)

Contention slots use slotted random access with carrier sense (listen-before-talk). Both modes use the same scheme with different timing. In the European duty-cycle bands carrier sense is not required by regulation but is always used.

Carrier sense: in the CCA window at the start of the slot (40 µs in HB, 200 µs in LR) the device measures the RSSI. The channel is busy above −85 dBm (HB) or −94 dBm (LR). If it is busy, the device selects a later slot in the same frame. After two failed retries the packet is dropped; the next fix follows within a second.

Carrier sense cannot detect a device that picked the same slot, because both listen at the same time. It detects other users of the channel and late transmissions from the preceding slot. The occupancy map avoids same-slot collisions.

Carrier sense override: persistent interference, for example from a nearby transmitter of another system, shall not silence a device. If carrier sense has blocked every transmission attempt of a device for 2 s, the device sends its next packets in its selected slots without carrier sense. It returns to normal carrier sense as soon as the channel is found free again. The override adds no traffic: the device keeps its current periodic rate and duty cycle limits. A device in distress (urgency 3) skips carrier sense after the first busy result. The override applies only where regulation permits it: in the European duty-cycle bands and with frequency hopping.

Occupancy map: every device marks a contention slot busy if it decoded a packet in it, or CCA reported it busy, in any of the last 4 occurrences of the same cycle phase. The cycle phase is the Unix second modulo 8, so devices reporting at 0.5, 0.25 or 0.125 Hz can share a slot on different cycles. The device draws its slot uniformly from the slots not marked busy; if none is free, from all contention slots of the frame.

Sticky slots: after a successful transmission without busy indication, a device reuses the slot in later cycles. It draws a new slot when the slot is found busy, and at random on average every 32 cycles to resolve conflicts between devices that cannot hear each other. Periodic traffic dominates, so the fleet converges to a nearly collision-free, TDMA-like pattern without coordination messages. Deviation reports are not periodic and always use a fresh draw in an HB contention slot.

A slot is free again when its occupant has not been heard in 4 consecutive occurrences of its cycle phase, so devices that leave radio range release their slots. Fleets out of range of each other reuse the same slots. The fourth-power path loss over water and FSK capture make this harmless at the receivers concerned.

Joining and rejoining: devices are switched on over 30 minutes to 3 hours before racing, in no particular order, and may be switched off and on again or lose reception for a while. A device that starts, restarts or regains reception listens for at least 8 s, one full round of the eight cycle phases, before its first periodic message, so that its occupancy map is filled; distress messages are exempt. It then draws a new slot; its previous slot may have been taken in the meantime and is released by the rule above. After a restart the device creates a new key chain and sends its chain commitment and certificate chain in its first authentication interval; receivers accept the new commitment because its anchor interval is later than the previous one.

Infrastructure slots, and RTK slots while RTK is announced, are not used for contention. They are assigned per venue in the configuration of the organizer's infrastructure (see Gateways). Devices shall not transmit outside the slot grid or across a frame boundary.

Preamble and sync word

Library: mandatory (obl_frame)

The preamble is 16 alternating bits, 0b1010…, ending with 0 (0xAAAA). It serves for AGC settling and bit synchronization and is not whitened.

The 3-byte sync word identifies OBL packets and follows the preamble (draft value):

	Byte 0	Byte 1	Byte 2
Hex	0xB4	0x2B	0x59
Binary	10110100	00101011	01011001

CRC

Library: mandatory (obl_frame)

The CRC is CRC-16/CCITT with polynomial 0x1021 and initial value 0xFFFF, computed over sync word, header and payload in transmission order and sent least significant byte first. Receivers shall discard packets with an invalid CRC. A reference implementation is given in Appendix C.

The CRC is independent of the FEC. At low SNR an iterative LDPC decoder can converge to a wrong codeword without noticing. The CRC detects such errors and serves as stop criterion for the decoder iterations.

Forward error correction

Library: mandatory (obl_frame)

OBL uses a systematic LDPC(192,144) code (Gallager code) with rate 3/4. The 144 data bits (header, payload, CRC) are sent unchanged, followed by 48 parity bits (6 bytes). The parity check matrix will be published as a machine-readable annex with the next revision.

Hard-decision decoding is the baseline, since most transceivers provide no soft bits. Soft-decision decoding is optional and gives a coding gain of about 4 dB in typical conditions. Receivers with little processing power may ignore the parity bits and rely on the CRC alone.

Under review: removing the FEC entirely and keeping only the CRC. Key arguments:

- The 48 parity bits take 21 % of every packet; without them, airtime drops accordingly and LR capacity rises by about 25 %.
- With hard-decision decoding, the net gain of this short code is estimated at only 0 to 1 dB, about 6 % range over water.
- Most losses in random access come from collisions, which FEC hardly helps.
- CRC-16 runs in hardware on practically all transceivers; without LDPC, devices need less computation and power.

The decision follows the planned capacity simulation and field measurements.

Data whitening

Library: mandatory (obl_frame)

Header, payload, CRC and FEC parity are whitened with a PN9 sequence ($x^9 + x^5 + 1$, seed 0x1FF). Whitening removes long runs of equal bits and replaces Manchester coding at full bit rate. Most transceivers provide PN9 in hardware, but the implementations differ in bit order and seed handling. An implementation shall either verify bit-exact agreement with the reference implementation (Appendix C) or whiten in software.

Coding methods

Exponential encoding

Library: mandatory (obl_coding)

Exponential encoding represents values with a large range in few bits. It works like a floating-point format but uses integer arithmetic only. The quantization step grows with the value, so the relative precision stays about constant: a boat drifting at 0.5 kn needs 0.1 kn resolution, a foiler at 45 kn does not. It is specified in the same form, including the sign convention, in EASA ADS-L 4 SRD-860, section G.1.6 (see References).

An exponentially encoded field has N code bits: Nm mantissa bits and Ne = N − Nm exponent bits. The code point c = (e << Nm) | m represents the value

$v = 2^e \cdot (2^{Nm} + m) - 2^{Nm}$

with a linear scaling factor given per field in the message tables. Decoding needs only shift, add and mask operations. The encoder selects the smallest of the 2^{Ne} exponents that can represent the value and rounds to the nearest code point. Exception: pos_acc rounds up, so that the transmitted value is an upper bound (see Position accuracy). Reference implementations are given in Appendix C.

Signed values, such as rate of turn, use the most significant bit as sign bit; the remaining N − 1 bits hold the magnitude, encoded as an unsigned value. Both encodings of zero are valid; senders shall use +0.

Relative coordinate encoding (RCE)

Library: mandatory (obl_coding)

A raw WGS-84 position needs 2 × 32 bits; RCE needs 40 bits at 4.44 cm resolution. It uses the fact that sender and receiver are within radio range of each other. Coordinates are sent relative to a grid much larger than the radio range, and the receiver selects the grid cell closest to its own position. The principle is the local decoding of Compact Position Reporting (CPR) in ADS-B (RTCA DO-260B).

Latitude and longitude are compressed separately, with the resolution divider D and the modulus m (a power of two):

$c_{comp} = \lfloor (c_{raw} + D/2) / D \rfloor \bmod m$

c_raw is in units of 1E-7 degrees. The division is floored, and mod is the mathematical modulo with non-negative result, as in Python (the C operator % differs for negative operands). The resolution is D × 1.11 cm and the grid width m × D × 1.11 cm, multiplied by cos(latitude) for longitude:

Dimension	D	m	Resolution	Grid width
Latitude	4	2^{20}	4.44 cm	46.6 km
Longitude, |lat| < 66°	4	2^{20}	4.44 cm	46.6 km × cos(lat) ⩾ 18.9 km
Longitude, 66° ⩽ |lat| < 79°	8	2^{20}	8.9 cm	93.2 km × cos(lat) ⩾ 17.8 km

One longitude divider covers all venues below 66° latitude; a single doubling covers higher latitudes. Latitudes of 79° and above are not supported.

Decoding

The receiver (1) decodes the latitude, choosing the grid cell closest to its reference latitude, (2) selects the longitude divider (4 or 8) from the decoded latitude, and (3) decodes the longitude in the same way. The reference is the receiver's own position, or a configured venue position for fixed installations. Decoding is unambiguous if sender and reference are less than half a grid width apart: at least 8.9 km, relay hops included. Race courses are well within this limit.

Elevated receivers, such as shore gateway nodes, can occasionally receive boats beyond this limit, in particular with ducting over water. The decoded position is then off by a whole grid width, with no error indication and a smooth track. To detect this, part 0 of the information message carries coarse_lat and coarse_lon, bits 18-23 of the scaled coordinates ⌊(c_raw + D/2) / D⌋. If they differ from the same bits of the decoded position by more than 1 (modulo 64), the receiver shall correct the position by the multiple of the grid width that makes them agree, or discard it.

Reporting

Reporting rates

Library: mandatory (obl_sched)

Tracking messages are sent in two ways:

- Periodic messages: sent in a sticky slot in the frames of the device's mode. They keep the slot pattern predictable and serve as keep-alive.
- Deviation reports: sent when the GNSS fix deviates from the prediction (see Extrapolation-based reporting). They are always sent in an HB contention slot of the next HB frame with a fresh draw, whatever the device's mode.

The periodic rate follows a binary ladder, aligned with the cycle phases used for slot sharing:

Rate	Transmission	Use
2 Hz	one packet in each frame of the device's mode	small fleets; distress
1 Hz	one packet per cycle	moderate load
0.5 Hz	one packet every second cycle	baseline
0.25 Hz	one packet every fourth cycle	high load
0.125 Hz	one packet every eighth cycle	overload

Rates above 2 Hz are not supported; extrapolation-based reporting makes them unnecessary. The frame and slot within the permitted cycles are chosen pseudo-randomly (see Medium access control).

The periodic rate adapts to the channel load. Each device takes the fraction of contention slots of its mode marked busy in its occupancy map as congestion measure. It uses the highest rate when this fraction is low and steps down as it rises. It steps down quickly and back up slowly, so the channel stays stable when the load rises quickly, as at a race start. The rate stays between 0.125 Hz and the maximum set by the event settings and the regulatory duty cycle. No signalling is needed: unused slots are released by the occupancy rule, and receivers bridge longer gaps by extrapolation. Typical operating points are given in Capacity and scaling.

Extrapolation-based reporting (EBR)

Library: mandatory (obl_ebr, prediction and report trigger); receiver display handling: recommended (obl_track)

Most of the time a boat moves as its last report predicts. With EBR, the transmitter extrapolates its own position from its last transmitted tracking message, compares each new GNSS fix with this prediction, and sends a deviation report if the horizontal deviation exceeds the threshold. This is dead reckoning with report-by-exception, as standardized for distributed interactive simulation in IEEE 1278.1. In its terms, the periodic message is the heartbeat and the deviation threshold is the dead-reckoning threshold.

The prediction assumes constant speed and course over ground: the last reported position is advanced along its COG at its SOG for the time since the fix. It shall be computed from the decoded values of the transmitted message, after quantization and coordinate reconstruction, not from internal full-precision values. Receivers therefore compute the same prediction as the transmitter.

The default deviation threshold depends on the device's performance tier: 0.2 m at tier 1, 1 m at tier 2 and 2 m at tier 3. It lies close to the fix-to-fix position noise of each GNSS class, so that reports carry real movement, not noise. A device also sends a deviation report when its heading differs from the heading of its last transmitted report by more than the heading threshold, 15° by default. Devices without a compass apply it to the course over ground, and only above 1 m/s. The heading threshold reports luffs and tacks earlier than the position threshold, which matters for umpiring; its default lies above the heading oscillation in waves. The race committee may set both thresholds in the event settings, and devices raise them under high channel load (see Capacity and scaling).

On straight legs boats send only periodic messages; tacks, gybes, mark roundings and starts trigger immediate reports. In the two cycles after a starting signal, boats send no deviation reports; the start report takes their place (see Race management message). This typically halves the tracking traffic or better. At the baseline rate, a new receiver acquires the fleet within two cycles.

Receivers may display the extrapolated position between reports; its error is bounded by the threshold plus the effect of lost packets. They extrapolate for at most three periodic intervals after the last report (the interval follows from the sender's periodic spacing, at most 8 s), then show the last position as stale with its age, so that a boat that has stopped transmitting, for example after a capsize, does not appear to sail on. They remove the boat from live displays after 10 minutes without reception; gateway nodes keep the last known state longer (see Gateways). Display software should blend a new report in smoothly, for example over 1 s; scoring, OCS identification and umpiring use the reported positions.

Relaying

Library: mandatory (obl_relay)

The relay field in the header holds the remaining hop budget. Boat nodes send tracking messages with relay = 0 by default; the race committee may request relay = 1 in the event settings. Messages with urgency 2 or 3 shall be sent with relay = 1.

Only relay nodes (umpire or rescue boats, marks, gateway nodes) retransmit. A relay node retransmits a packet only if its relay field is not zero and it has not already received a relayed copy. It decrements the relay field, leaves all other bytes unchanged and transmits in a random contention slot of the next frame of the same mode; relay nodes may use infrastructure slots. Receivers remove duplicates by address, message type and time field. One hop roughly doubles the covered radius. The RCE range limit applies to the total distance including the hop.

On-board recording

Library: recommended (obl_log)

Tier 1 and 2 devices shall record every GNSS fix on board: time, position, SOG, COG, heading and GNSS receiver status including RTK, each value with its accuracy estimate. The record is kept for at least 36 hours, so that the previous day can always be read, or until the race committee has read it out. The device interface, or a publicly available conversion program, shall export the record as CSV with all fields; the reference library provides such an export. Storage is small: 10 Hz over an 8-hour race day is about 10 MB.

Recording and transmission are separate. The radio carries tracking at the rates of this chapter, which the channel capacity sets. Start and finish decisions use the on-board record and the dedicated reports (see Race management message), not the transmitted track. Tier 3 devices need no record.

Addressing

Every device has a static 24-bit radio address. Addresses shall be unique on the water; duplicate addresses corrupt tracking, umpiring and safety functions.

Range	Use
0x000000	Invalid, never transmitted
0x000001 – 0xBFFFFF	Boat devices; blocks assigned to manufacturers and national authorities by the OBL registry
0xC00000 – 0xDFFFFF	Infrastructure nodes: committee, mark, relay, gateway and RTK base nodes
0xE00000 – 0xEFFFFF	Development and vendor experimentation
0xF00000 – 0xFFFFFF	Test range; shall not be used at events

The information message links an address to sail number, class and name; event software maps addresses to entries at registration. OBL uses no randomized privacy addresses, since racing is public. The no_track flag covers training and transit.

Authentication

Library: mandatory (obl_auth), all sections of this chapter

This chapter is a draft. OBL protects messages against forgery, alteration and replay. Encryption of boat data is an event option (see Encryption). Every receiver, including other boats, can check a message on its own, without shared secrets, online infrastructure or keys per event.

Building blocks

OBL combines asymmetric and symmetric cryptography:

	Asymmetric (public-key) cryptography	Symmetric cryptography
Keys	Key pair: a private key that stays in the device and a public key that is published	One secret key, known to sender and verifier
Algorithms in OBL	ECDSA on curve P-256 with SHA-256	HMAC-SHA-256, truncated to 24 bits; SHA-256 hash chain

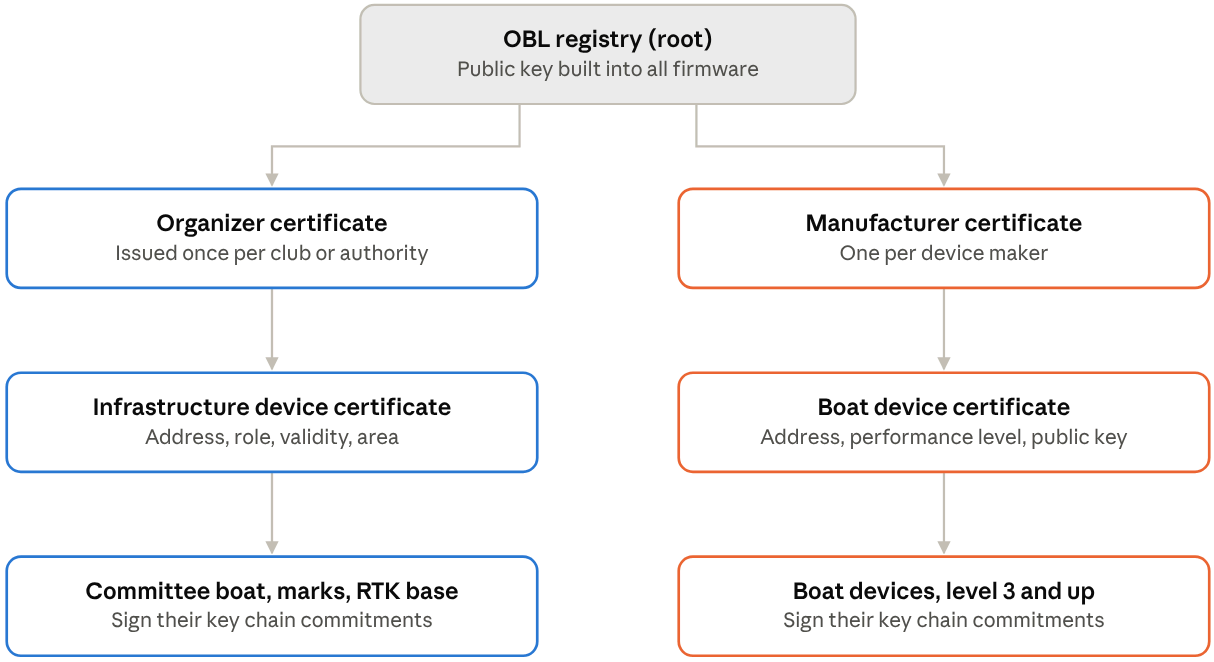
	Asymmetric (public-key) cryptography	Symmetric cryptography
Proves	The signer holds the private key; anyone with the public key can check	The sender knew the key when it sent the packet
Size on air	512-bit signature, spread over several packets	24-bit MAC in each authentication message
Computation	High, done rarely	Low, done for every packet
Used for	Certificates and key chain commitments	Protecting each packet

A symmetric MAC alone does not work for a broadcast: every receiver would need the key and could then forge packets itself. OBL therefore keeps each symmetric key secret while it is in use and publishes it afterwards, when it can no longer be used for forgery (delayed key disclosure). Asymmetric signatures tie each key chain to its device, and certificates tie the device to a trusted issuer. P-256 is used because secure elements and MCU-internal key stores of all major manufacturers support it; Ed25519 is not widely supported.

Trust chain

The trust chain is a small public-key infrastructure with one root and two branches: organizers certify their infrastructure, manufacturers certify their boat devices. Each certificate is signed with the private key of the level above it, and a receiver checks the chain from the device certificate up to the root.

Every certificate leads back to the OBL registry



Each arrow: the key above signs the certificate below. Organizer branch left, manufacturer branch right.

OBL trust chain · one root, two branches

- Root: the public key of the OBL registry, built into the firmware of every device. It is the only trust information a device needs.
- Organizer certificate: the registry, or a national authority it delegates to, certifies the public key of an organizing authority. It is issued once, when the club or organizing authority registers, and renewed automatically.
- Manufacturer certificate: the registry certifies the public key of each device manufacturer with an assigned address block.
- Infrastructure device certificate: the organizer certifies each of its infrastructure devices with address, role (race committee, mark, RTK base, gateway, relay), validity period and area. The organizer chooses validity and area: the days and waters of an event, or a season and the club's home waters, so that recurring club races need no certificate handling. The organizer certifies its encryption public key in the same way (see Encryption).
- Boat device certificate: the manufacturer certifies each device of tier 1 or 2 at production, with its address, performance tier and public key.

Each device generates its key pair in its key store (see Key store); the private key never leaves it. An extracted key allows forging only that one address, and the race committee can exclude it. Devices send their certificate chain in authentication messages (msg_type 7, subtype 1) at least every 5 minutes; receivers cache verified certificates. No list of keys has to be distributed. The certificate format is still to be defined.

Receivers apply these rules:

- Race management messages, event settings, RTK corrections and mark commands are accepted only from infrastructure devices whose chain leads to the root and whose role, validity and area match the message, the GNSS time and the receiver's position.
- Boat data counts as verified only if the chain of the boat device leads to the root, the address in its certificate matches the packet, and the certified performance tier is not lower than the tier the device declares.
- Limited validity makes revocation lists unnecessary for infrastructure. If a long-lived infrastructure device is lost, the organizer renews its own key and re-certifies its remaining devices.
- Data without a valid chain, including all data from tier 3 devices, may be displayed as not verified but shall not be used for automatic decisions.

Key store

Every authenticating device keeps its private key in a key store: a secure element, or a key store inside the MCU. A key store shall provide at least:

- Key generation on the device with a hardware random number generator, at the latest during production.
- No read access to the private key after generation, for firmware, debug interfaces and test modes. Firmware can only request signatures and key agreement.
- ECDSA P-256 signing and ECDH P-256 key agreement inside the key store, or in crypto hardware that receives the key directly from it.
- Countermeasures against physical attacks: detection of voltage and clock glitches, and side-channel protection of the signing operation.
- Debug access to the key store locked permanently in production devices.

An MCU-internal key store qualifies only if hardware separates the key from the application firmware. A key in ordinary flash memory, protected only by readback protection, does not qualify.

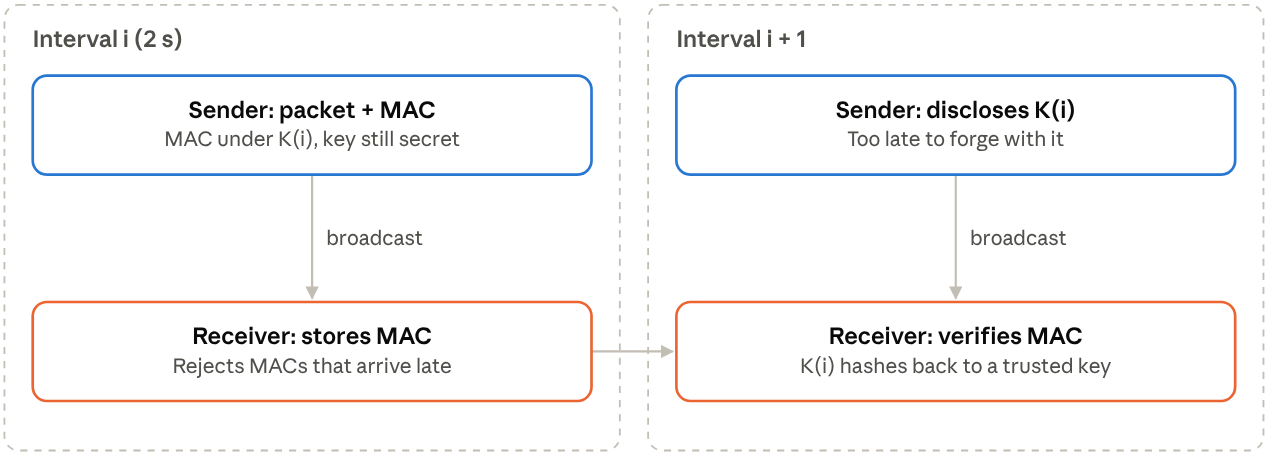
Registration

The trust chain proves that a device is genuine; it does not say which boat carries it. At registration for an event, each entry names the addresses of its devices. The race committee accepts data only from registered addresses and flags a device whose information message (sail number, class) does not match its entry.

Per-packet authentication

A packet cannot carry a 512-bit signature. OBL therefore protects each packet with a short symmetric MAC and discloses the key one interval later, following TESLA (RFC 4082). The GNSS time base provides the synchronized clock that TESLA needs.

A key is disclosed only after it can no longer be used to forge



Delayed key disclosure · two intervals

Step 1, key chain. At power-up the device draws a random 56-bit key K(n) for its last interval n and computes the chain back to the anchor interval a:

```
K(i-1) = trunc56(SHA-256(0x01 || address || i || K(i)))
```

Address and interval index in each step make all chains independent and prevent precomputed tables. A 24-hour chain has 43 200 elements and is computed in well under a second on a typical microcontroller.

Step 2, commitment. The device signs address ‖ a ‖ K(a) with its private key and broadcasts this chain commitment (msg_type 7, subtype 1) at least every 60 s.

Step 3, MAC. For each packet sent in interval i, the device computes

```
mac = trunc24(HMAC-SHA-256(trunc128(SHA-256(0x02 || K(i))), header || payload))
```

and sends it in its authentication message of that interval.

Step 4, disclosure. In interval i + 1 the device discloses K(i).

Step 5, verification. The receiver checks that repeated hashing of K(i) leads back to a key it already trusts, and then checks the stored MAC. Lost disclosures only delay verification, since any later key also leads back. A lost authentication message leaves its own packet unverified and nothing else; because tracking is periodic and race management messages are repeated, a loss costs at most one interval or one repetition of verified data.

Intervals and timing

Interval i of a device covers the Unix seconds 2i + p and 2i + p + 1, where p is the least significant bit of its address. The alternating phase preserves slot sharing between even and odd cycles. A device that authenticates its periodic tracking sends the tracking message in the first second of the interval and the authentication message in an HB frame of the second second, at the latest in F1.

A receiver accepts a MAC for interval i only if the authentication message arrives before interval i ends by its own clock; later copies, including relayed ones, are discarded. This rule prevents forgery, since the key of interval i is disclosed only in interval i + 1. If a receiver gets two different MACs for the same packet, the packet counts as not authenticated.

Scope

- Infrastructure nodes (committee, mark, gateway and relay nodes) authenticate every race management, safety and weather message it originates, including mark commands. Robotic marks shall execute only verified commands.
- RTK: the RTK base node appends a 24-bit MAC to each RTCM message before fragmentation, using the key of the interval of the last fragment, and discloses its keys with subtype 2. Receivers pass corrections to the GNSS receiver as soon as they arrive, without waiting for verification. If verification later fails, the device marks the positions computed with those corrections as invalid (rtk_fix cleared, pos_acc = 0) until corrections verify again. Forged corrections are rare and detected by every receiver; outdated firmware or keys are the usual cause of failures.

- Boat nodes authenticate in the mode set in the event settings, default mode 1. Mode 0: none. Mode 1: periodic tracking and information messages, and start and finish reports. Mode 2: all messages. In mode 1 deviation reports are not authenticated; a forged report is shown at most until the next authenticated periodic message and is never used for scoring or protests. Tier 3 devices always use mode 0. Start and finish reports (msg_type 4, subtypes 8 and 9) are authenticated in modes 1 and 2 and never encrypted; the committee uses them as received for the start result and treats a report that fails verification as missing.
- Distress alerts are never suppressed for lack of verification. Receivers alert their operators and mark the alert as unverified.

Use of verified data

Receivers display data immediately and mark it as confirmed once verified, typically 1-3 s later; race management messages may likewise be displayed before verification, marked as unconfirmed. With authentication enabled, scoring, automated OCS and finish detection, and data used as evidence in protests shall use verified data only.

Boat nodes send authentication messages in HB contention slots, infrastructure nodes in their infrastructure slots. Authentication messages of low-antenna devices reach nearby boats and elevated receivers; receivers beyond HB range see such data as unconfirmed unless a relay node forwards the authentication messages. The channel load is part of the capacity budget.

Encryption

Library: mandatory (obl_auth), all sections of this chapter

This chapter is a draft. OBL encrypts the live data of boats when the event requires it. Two purposes: boats cannot use each other's positions while racing, and the organizer, or a class on its sailors' behalf, controls who distributes live tracking data, for example as a media asset. Third-party receivers near the course see only encrypted packets. The race committee, umpires and authorized gateways read the data live; everyone else can read it once the organizer publishes the keys. Encryption uses the device keys and the trust chain of the Authentication chapter and adds no bits to the packet.

What is encrypted

With enc = 1 in the event settings, boat nodes encrypt their tracking messages (msg_type 0) and telemetry messages (msg_type 2). The header and the ver and time fields stay clear, and for telemetry also subtype; all following payload bits are encrypted. Relay nodes forward encrypted packets unchanged.

All other messages stay clear, since every boat must read them: information, safety and distress messages of boats, and all messages of infrastructure nodes, including mark positions. Start and finish reports stay clear (see Race management message).

enc = 1 requires auth_mode 1 or 2. Tier 3 devices have no key store and send in clear, so events that encrypt should require tier 2 or 1.

Keys

Each boat device and the organizer derive a key that only the two of them know, from keys that already exist. The chain keys of delayed key disclosure are not suitable, since they are published 2 s after use.

1. The organizer has an encryption key pair on P-256. It certifies the public key with validity period and area, like an infrastructure device, and its infrastructure nodes send this certificate with their certificate chain.
2. The boat device takes the encryption key of the organizer whose event settings it follows. Its key store computes the shared secret Z by ECDH from the device private key and the organizer's encryption public key.
3. The organizer computes the same Z from its encryption private key and the public key in the boat device certificate.
4. Both derive one key per boat and UTC day d (Unix time / 86 400):

```
K(d) = HKDF-SHA-256(Z, info = "OBL-ENC" || address || d), first 16 bytes
```

No key exchange takes place on air and no configuration is needed. The organizer installs its encryption private key, in a key store, on the receivers that need live data (committee, umpire and gateway nodes), or gives them only the day keys.

An opened boat device reveals only its own data. A lost encryption private key of the organizer reveals the data within its validity, so the organizer should keep the validity short: an event, or a season for club racing.

Cipher

AES-128 in counter mode (CTR). The counter block is unique per packet:

```
ctr = address (24) || msg_type (3) || subtype (5; 0 for tracking) || t (40) || 0
(56)
```

t is the Unix time of the packet in tenths of a second, restored from its time field (see Field semantics). The encrypted bits are XORed with the first bits of AES-128(K(d), ctr). A sender shall not send two packets with different content and the same counter block. One AES block per packet is needed; common sub-GHz SoCs have AES hardware.

The MAC of per-packet authentication covers the packet as transmitted, so every receiver, including other boats, can still check that an encrypted packet is genuine.

Receivers

A receiver with the key decrypts a boat's packets once it has the boat device certificate, which is sent at least every 5 minutes; it stores earlier packets and decrypts them then. Other receivers cannot decode encrypted packets and may store them for later.

Key publication

The organizer decides when it publishes the day keys of all boats: after the last race of the day, after the event, or later, at the time given in the sailing instructions. It shall give the protest committee and the parties to a protest access to the data concerned. Publication is recommended: anyone who recorded the packets can then decrypt them, so all data becomes public and auditable. Public live tracking on the internet must be delayed until then, otherwise boats can get the data by phone; this is a matter for the backend and the event.

Open points: access of coaches to the data of their own boats; certificate format of the encryption key.

Capacity and scaling

OBL is dimensioned for full performance with up to 150 boats within radio range, which covers almost all regattas, and degrades gracefully up to 600 boats. Nearby course areas share the LR capacity; with separate HB channel sets they do not share the HB capacity (see Channel plan).

Design point and degradation

- Up to 150 boats: displayed positions stay within the deviation threshold 99 % of the time, manoeuvres are reported within 1 s, and authentication and telemetry are available.
- 150 to 600 boats: position errors and reporting delays grow step by step; every boat sends at least one periodic report every 8 s.
- More than 600 boats: no guarantee, but no collapse. Periodic positions, race management and safety traffic continue.

Under rising load, each device degrades in a fixed order, based on the load it measures itself, and reverses the steps as the load drops: (1) telemetry is switched off, (2) boat node authentication is reduced to mode 1, (3) the deviation and heading thresholds are raised, (4) the periodic rate is lowered to 0.25 Hz and then 0.125 Hz. Infrastructure slots (race management, relay and gateway nodes), active RTK streams, safety messages and distress alerts are never degraded.

Capacity budget

Quantity	HB frames (F0, F1)	LR frames (F2, F3)
Raw bit rate	300 kbit/s	62.5 kbit/s
Packet on-air time	773 µs (232 bits) in a 1 ms slot	3712 µs (232 bits) in a 4250 µs slot
Slots	250 × 1 ms per 250 ms frame	58 × 4.25 ms per 250 ms frame
Slot instances per second (2 frames)	500	116
Infrastructure slots	14 per second (slots 1-7; slot 0 is the retuning gap)	8 per second (slots 0-3); plus 16 RTK slots per second (slots 4-11) when RTK is active
Contention slots	484 per second	108 per second (92 with RTK)
Periodic tracking capacity at 0.5 Hz (sticky slots, slot sharing)	≈ 970 devices; ≈ 780 recommended	≈ 216 devices; ≈ 170 recommended (184 / 145 with RTK); ≈ 860 at 0.125 Hz

Quantity	HB frames (F0, F1)	LR frames (F2, F3)
Contention load at a race start, whole fleet manoeuvring (150 boats, ≈ 1 deviation report/s each, authentication mode 1)	≈ 55 % of contention slots	0 (deviation reports use HB)
Collision behaviour	deviation reports: ≈ 42 % at the race-start peak, ≈ 25 % effective after capture (to be confirmed by simulation); ≈ 15 % in steady state	periodic messages: collision-free after sticky-slot convergence
Probability of no usable update from a manoeuvring boat for > 3 s	≈ 2 %	-

Periodic tracking in sticky slots is collision-free after convergence and is treated as scheduled traffic; a venue is dimensioned by the periodic capacity row. Deviation reports contend randomly in HB frames. Their load peaks when the fleet manoeuvres together, at race starts and crowded mark roundings. Switching devices on is not a peak: it is spread over 30 minutes to 3 hours, and joining devices listen before they transmit (see Medium access control). The burst row assumes that each of 150 boats sends one deviation report per second, plus authentication in mode 1: about 0.7 HB packets per second per device (one authentication message per 2 s, seven commitment packets per minute and the certificate chain every 5 minutes), or about 105 packets per second, 22 % of the HB contention slots.

A lost deviation report is retried at the next fix, receivers extrapolate meanwhile, and elevated gateway nodes receive close to 100 % of the traffic.

At a start, the start reports of the fleet replace the deviation reports in cycles S+1 and S+2: 150 reports in 484 HB contention slots, about 30 % load, each sent twice, so about 95 % arrive (to be confirmed by simulation; see Race management message).

Simulation scenario for Olympic-class events: three fleets of 60 boats on course areas 1.5 km apart start at the same second, with start reports, with and without separate HB channel sets. The results are required before the protocol is adopted for such events.

Operating points

Devices choose their operating point from the load they measure; the event settings can only lower the maximum rates. Typical operating points, at a contention load of about 30 % or less outside race starts:

Boats within radio range	Periodic rate (HB / LR installations)	Telemetry
⩽ 20 (match race, medal race)	2 Hz / 2 Hz	up to 5 Hz
⩽ 150	1 Hz / 0.5 Hz	up to 2 Hz
⩽ 300	0.5 Hz / 0.25 Hz	disabled
⩽ 600	0.25 Hz / 0.125 Hz, raised deviation threshold	disabled

Regulatory duty cycle

In Europe, each device shall track its own transmit time per band over a rolling hour and stay within the limit of each band: 1 % in the HB band and 10 % in the LR band. When it approaches a limit, it reduces optional traffic in the degradation order (see Design point and degradation). Typical values per device:

- LR band: periodic tracking at 0.5 Hz uses 0.19 %, at 2 Hz (distress) 0.74 %.
- HB band: deviation reports (up to 1 per second) use 0.08 %, authentication 0.05 %, periodic HB tracking at 2 Hz 0.15 %. Telemetry uses 0.08 % per Hz, 0.4 % at its maximum of 5 Hz.
- RTK base node: about 5.5 % in the LR band (see RTK correction message).

In the USA and Canada, frequency hopping under FCC §15.247 has no duty cycle limit; the dwell time rule is met because a device transmits at most a few milliseconds per 250 ms frame.

Gateways

Library: recommended (obl_gw)

OBL to internet

A gateway node receives OBL traffic and forwards it to the internet. It follows the channel sequence with one receiver, or receives HB and LR channels in parallel with two. Each packet is forwarded with reception metadata: gateway identifier, timestamp (nanosecond resolution where available), RSSI, estimated SNR and CRC status. Transport is UDP or MQTT with compact CBOR encoding; a JSON representation is defined for debugging. The backend removes duplicates received by several gateway nodes using (address, msg_type, time).

For search and rescue, every gateway node shall keep the last known state of each address it has received for at least 48 hours, also across restarts: the latest tracking position with timestamp, the information message contents (identity, MMSI, boat type, dimensions) and the safety check-in state. The record shall be queryable locally, so that SAR has a last-seen picture even when a boat has stopped transmitting or the internet link is down.

Typical gateway sites are the committee boat, coach boats and one or two elevated shore stations. Clubs can add coverage with low-cost receivers: a transceiver module on a small Linux board or a microcontroller with network access. The backend merges all gateway nodes into one live picture with sub-second latency.

Internet to OBL

Transmit-capable gateway nodes inject traffic from shore: race management messages from scoring software and RTK streams from a base station ashore or from a network correction service (see RTK correction message). This traffic is restricted to infrastructure and RTK slots and to message types 3, 4, 5, 6 and 7 (RTK corrections, race management, safety acknowledgements from shore, weather from land stations, and their authentication messages). It never uses contention slots. The organizer assigns these slots to its transmitting infrastructure nodes when configuring them, so infrastructure traffic is collision-free.

Backend services

Aggregation, storage, live distribution, replay, umpiring feeds and public APIs are backend functions outside this specification. The protocol allows anyone to build them from any combination of gateways.

Hardware implementation

Every component shall be a common part available from at least two manufacturers. The protocol therefore uses only functions that all such parts provide:

- Transceiver: raw packet mode, RSSI carrier sense and the bit rates of both modes. Whitening, CRC and LDPC can be computed in software on the MCU.
- GNSS receiver: time pulse output (e.g. u-blox, Quectel).
- Key store (tiers 1 and 2): secure element (e.g. Microchip, NXP, Infineon, ST) or MCU-internal key store, with ECDSA P-256 and ECDH (see Key store and Encryption).
- Heading sensor (tiers 1 and 2): three-axis magnetometer with three-axis IMU (gyro and accelerometer), or dual-antenna GNSS with antennas 20 cm or more apart.

Radio transceiver

Library: recommended (obl_hal)

Suitable are sub-GHz transceivers and modules with 2-GFSK at 300 kbit/s, for example Semtech SX126x and SX127x, Texas Instruments CC1200 and CC13xx, Silicon Labs Si446x and EFR32, and ST S2-LP. Pre-certified modules (FCC/CE) cost about USD 2-5 at 1000 pieces and remove most of the certification effort (see Radio certification). Required are programmable deviation and bandwidth, RSSI carrier sense and a raw packet mode. Soft-decision output is optional.

GNSS

Tier 3 accepts any GNSS module with at least 1 Hz. Tier 2 needs a multi-constellation module at 10 Hz, single- or multi-band (e.g. u-blox M10 class, USD 2-10). Tier 1 needs an RTK receiver at 10 Hz that uses the venue RTCM stream (e.g. u-blox F9/X20 class). A good antenna with clear sky view and a ground plane matters more than the receiver grade. The time pulse output is required at all tiers for slot timing (±5 µs).

Antenna and installation

A vertical quarter-wave whip, mounted as high and as clear of water, crew and rigging as possible, is sufficient. On dinghies the mast top or the transom flagpole position works well. Devices worn on the body or mounted at deck level lose several dB through waves and hulls. The link budget tolerates this at short range, but relays and gateways should avoid it.

Radio certification

A module certification (CE, FCC, ISED) covers only the tested configuration: the firmware with its RF parameters, and the listed antennas. The OBL PHY (2-GFSK at 300 and 62.5 kbit/s, OBL channel plan, frequency hopping in the USA, Australia and NZ) is usually not part of a vendor's test reports. OBL therefore defines a shared certification path:

1. The editor group selects one or two reference modules per region, for example an SX1262 transceiver module and an STM32WL module.
2. The module vendor or the editor group extends their certification with the OBL PHY and channel plan: in Europe with RED test reports to EN 300 220, in the USA with a Class II permissive change of the modular grant.
3. Grants and test reports are published with the reference code.
4. Manufacturers that use a reference module, the official radio library unmodified and an antenna within the limits below reuse these results. They still declare conformity for their product (EU declaration of conformity, FCC Part 15B unintentional emissions, labelling).

With transceiver modules without own firmware, the host software sets the RF parameters. Their certification then holds only together with the mandatory radio library.

Antenna gain is limited:

- Europe: the limits are ERP. A device shall reduce its conducted power by any antenna gain above 2.15 dBi (0 dBd). A 14 dBm module with an antenna of at most 2.15 dBi meets the 25 mW ERP limit.
- USA: the grant lists antenna types and their maximum gain. An antenna of the same type with equal or lower gain needs no new tests (FCC 15.204). User-replaceable antennas need a non-standard connector such as RP-SMA (FCC 15.203).

```
P_{tx} \le P_{ERP,max} - \max(0,\ G_{ant} - 2.15\,\mathrm{dBi})
```

Reference certifications use a vertical antenna of at most 2.15 dBi. Boat antennas of 9 cm or less stay well below this gain; the limit matters for infrastructure nodes with collinear antennas.

Funding and holder of the reference certifications: see Governance.

Power and cost

Average power is dominated by the GNSS receiver (about 30-100 mW), not by the transmitter (25 mW peak at 0.2 % duty cycle). A node with one 18650 cell runs for several race days. The complete electronics of a meter-class standalone node (transceiver, GNSS module, antennas, MCU, battery charging) cost USD 20-30 in bill of materials. For comparison, a retail tracker board of this class (nRF52840, SX1262, GNSS, battery), the Seeed Wio Tracker L1, sells for about USD 31. Integrated into a device that already has GNSS, OBL adds less than USD 10 (see Appendix B). RTK devices are dominated by the GNSS receiver cost: a multi-band RTK module and antenna add about USD 100-150 at 2026 prices.

Conformance disclosure

Every product or software stack that claims OBL support shall publish an Implementation Conformance Statement (ICS). The ICS names the specification version, the firmware or software version and the release of the official library it uses. It assigns a status to each subtopic, that is, to each subsection of this document, with each message type counted separately. It shall be public and kept current. The ICS also determines which OBL performance tier a device may declare (see OBL performance tiers).

- Library: implemented by a mandatory module of the official library, used unchanged. This is the only valid status for subtopics that a mandatory module covers.
- Complete: implemented by the product, with own code or a recommended module, as specified and verified by test.
- Modified: implemented with deviations, each described in the ICS. Not permitted for subtopics of mandatory modules.
- Partial: only a subset is implemented; the subset is described in the ICS.
- Not implemented: the subtopic is absent.

Example excerpt of an ICS:

Subtopic	Status	Notes
Modulation and modes	Complete	SX1262, obl_hal driver
Medium access control	Library	obl_sched, release 1.0.2
Extrapolation-based reporting (EBR)	Library	obl_ebr, release 1.0.2

Subtopic	Status	Notes
Flags (hdg_valid)	Modified	magnetometer only, gyro check not implemented
Telemetry message (msg_type 2)	Partial	subtype 4 only
RTK correction message (msg_type 3)	Not implemented	tier 2 device

A published, accurate and current ICS is a condition for using the OBL name and logo. Products without an ICS, or with an inaccurate one, shall not be marketed as OBL or OBL-compatible. Subtopics marked Modified or Partial shall not be advertised as conformant. The subtopics that govern the shared channel and authentication are covered by mandatory modules and therefore carry the status Library in every transmitting product; a product that cannot declare this shall not transmit as OBL.

Reference library

The official OBL library implements the core of this specification. Its modules are either mandatory or recommended:

Module	Covers	Class
obl_frame	Packet assembly: preamble and sync word, bit packing of header and all message layouts, CRC, FEC, whitening	Mandatory
obl_coding	Exponential encoding, relative coordinate encoding including the aliasing check	Mandatory
obl_sched	Channel plan and hopping sequences, frame and slot timing, carrier sense decisions, occupancy map, sticky slots, override, joining, rate ladder and adaptation, duty-cycle accounting	Mandatory

Module	Covers	Class
obl_ebr	Prediction model and report trigger of extrapolation-based reporting	Mandatory
obl_relay	Relay decision and de-duplication	Mandatory
obl_auth	Key chain, MAC, interval timing, commitments, certificate chain verification, key store interface, encryption	Mandatory
obl_rtk	RTCM fragmentation and reassembly including the MAC trailer	Mandatory
obl_geo	Reduction of the antenna position to bow and hull outline, including heel and pitch compensation; magnetic declination from the World Magnetic Model	Mandatory
obl_rc	Race management application: countdown, start report, start result and review of missing reports, finish detection	Recommended; mandatory for systems used as ARMS Category 2
obl_hdg	Heading fusion of magnetometer and IMU, validity check	Recommended
obl_track	Receiver-side tracks: extrapolation for display, stale timeout, smoothing	Recommended
obl_log	On-board record of GNSS fixes and CSV export	Recommended
obl_hal	Drivers for the reference transceivers, PPS capture, GNSS parsing	Recommended
obl_gw	Gateway encoding (CBOR, JSON) and transport	Recommended
Python tools	Simulation, decoding, test vectors, conformance tests	Recommended

Rules:

- Products shall use the mandatory modules unchanged, from a released library version. The ICS names that version, and the firmware reports the version and the module checksums through its diagnostic interface.
- The conformance tests of the mandatory modules are run once per library release. A product that uses them unchanged is tested only for radio hardware, timing accuracy and integration.
- A product that modifies or replaces a mandatory module is not conformant for the subtopics the module covers. Since the shared-channel subtopics must be Complete for every transmitter, such a product shall not transmit as OBL.
- Recommended modules may be replaced by own implementations that pass the conformance tests for the sections concerned.
- Mandatory modules change only with a new specification version. Products update to the new release within the transition period that the release notes state.

FAQ

Q: Why 868/915 MHz and not 2.4 GHz?

A: Sub-GHz signals propagate much better over water: lower path loss, better diffraction around waves and hulls, less absorption by spray and rain. The 2.4 GHz band is crowded with Wi-Fi and Bluetooth at regatta venues. Sub-GHz antennas are still small enough for buoys and dinghies.

Q: Why not LoRa?

A: Airtime. Even at the fastest useful settings a LoRa position report takes many milliseconds, so 150 boats reporting every second do not fit, and frequent transmissions conflict with duty-cycle limits. GFSK at 300 kbit/s gives up a few dB of sensitivity for 10-20 times the capacity.

Q: Why slotted random access with listen-before-talk, and not TDMA?

A: It is simple, needs no registration or slot assignment for competitors, degrades gracefully under overload and is easy to make interoperable. The slot grid, based on the GNSS timing every device has, roughly doubles the throughput compared to unslotted ALOHA, and LBT resolves most remaining conflicts. Infrastructure that needs deterministic access (RTK, race management) has scheduled slots.

Q: Why no mesh?

A: With moving nodes a mesh cannot guarantee coverage, and it adds overhead and complexity. OBL uses medium-range links with an optional single relay hop. This extends coverage and keeps the protocol stateless and predictable.

Q: Why not AIS?

A: AIS is safety-of-navigation infrastructure on protected VHF channels. Its 9.6 kbit/s SOTDMA channels cannot carry 150 targets at racing update rates, transponders are expensive, and regulators do not allow sports traffic on the band. Boats that must carry AIS keep it.

Q: Why not cellular or satellite trackers?

A: Cellular coverage on the water is often marginal and drops when spectators load the network. The modem dominates power consumption and cost, every device needs a subscription, and all data passes through a server, which adds latency. And there is no boat-to-boat path, which umpires, race committee and safety boats need. OBL gateways connect to the internet where coverage exists.

Q: Is OBL encrypted?

A: Optionally. All critical data is authenticated: race management, RTK and, where the event requires it, tracking messages carry MACs that every receiver, including other boats, can verify (see Authentication). An event can also encrypt the tracking and telemetry of boats, so that boats cannot use each other's live data while racing. The race committee, umpires and authorized gateways read it live; the organizer publishes the keys after racing, so all data becomes public (see Encryption). Encryption also lets the organizer or a class control who distributes live tracking data. The no_track flag covers privacy outside competition.

Q: Can OBL replace flags and hails under the racing rules?

A: Yes, if the sailing instructions change the rules concerned, as events using an Augmented Race Management System (ARMS) already do. Appendix D gives sailing instruction text.

Q: What does an OBL device cost?

A: It depends on the tier, mainly through the GNSS receiver and antenna. A standalone Tier 3 node starts at about USD 20 in parts, a Tier 2 node with a 10 Hz multi-constellation module at USD 25-35. A Tier 1 node needs an RTK receiver and a multi-band antenna, about USD 100-150 more at 2026 prices. Adding an OBL RF module to a device that already has GNSS costs about USD 8; see Hardware implementation and Appendix B.

Governance

This chapter is a proposal. The registry holds the root key, assigns address blocks and certifies manufacturers and organizers (see Trust chain). Who runs it, who funds it and who it answers to is settled as follows, unless World Sailing chooses to govern the registry itself.

- Draft phase, until World Sailing recognition: a suitable national authority operates the registry and holds the root key on behalf of the editor group.
- Afterwards: a non-profit OBL association, open to national authorities, classes and manufacturers, takes over the registry, the root key and the reference radio certifications. The editor group becomes its technical committee.
- Root key: generated and kept in a hardware security module, with two members of the editor group present at the ceremony. A published rollover procedure replaces it; firmware carries the current and the next root key.
- Funding: a one-off fee per address block from manufacturers. Organizer certificates are free of charge for national authorities and their clubs. The reference certifications (see Radio certification) are paid from the fees, the first round by that national authority.

Contributing

OBL is developed in the open, and review, comments and proposals are welcome. An editor group led by Swiss Sailing decides which changes are adopted and keeps the protocol coherent, implementable and backward compatible. The specification is submitted to World Sailing as open protocol for Augmented Race Management Systems (see Appendix D).

During the draft phase this document is the master copy. Please comment directly in the document, or send feedback by e-mail to the editors, naming the section concerned. Wording proposals, problem reports and measurements from the water are all welcome.

Once the draft is stable, the project moves to a public GitHub repository. The specification source will be converted to Markdown, and PDF and HTML versions will be generated automatically. Defects are then reported as issues, questions and ideas as discussions, and wording changes as pull requests. Only the editors merge into the protected main branch, and each accepted revision is tagged as a release with its PDF.

The specification text will be published under CC BY 4.0 and the official C libraries and the reference implementations of Appendix C under a permissive open-source licence (MIT), so that commercial and open implementations can adopt OBL freely.

References

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- US Code of Federal Regulations, 47 CFR §15.247: Operation within the bands 902-928 MHz, 2400-2483.5 MHz and 5725-5850 MHz.
- RTCM Standard 10403: Differential GNSS Services, Version 3.
- NOAA NCEI and BGS: World Magnetic Model (WMM), current edition.
- World Sailing: The Racing Rules of Sailing 2025-2028.
- World Sailing Policy D12: Recognition Framework for Augmented Race Management Systems (ARMS), 27 February 2026.

Appendix A: Message definitions

Library: mandatory (obl_frame, all message layouts)

A packet carries a 4-byte header and a 12-byte (96-bit) payload specific to the message type. Senders shall set reserved fields to 0; receivers shall ignore them.

Header

Field	Width	Pos	Description
address	24	0	OBL radio address of the sender (see Addressing)
msg_type	3	24	Message type, see table below
urgency	2	27	Urgency level, see Field semantics
relay	2	29	Remaining relay hop budget; decremented by relays; 0 = do not relay
reserved	1	31	Set to 0

Message types

msg_type	Message	Typical sender	Min rate	Max rate
0	Tracking	Boat nodes	0.125 Hz	2 Hz
1	Information	Boat node	every 15 s	every 15 s
2	Telemetry	Boat node	optional	5 Hz (event settings)
3	RTK correction	RTK base node, RTK slots	14 packets/s	16 packets/s
4	Race management	Committee node; boat nodes for reports (subtypes 8-15)	every 10 s	1 Hz
5	Safety	Boat nodes	on event	1 Hz

msg_type	Message	Typical sender	Min rate	Max rate
6	Weather	Mark, committee and land-station nodes	every 30 s	every 10 s
7	Authentication	Authenticating nodes	once per 2 s interval with authenticated traffic	one per authenticated packet

A node shall meet the minimum rate and shall not exceed the maximum rate. Event-driven and optional messages have no minimum rate.

Tracking message (msg_type 0)

The tracking message carries everything needed for tracking, race management and umpiring in one payload. It is sent periodically and as deviation report (see Reporting).

Field	Width	Pos	Scaling, unit, range	Description
ver	3	0	0-7	Protocol version. This document describes version 1.
time	7	3	0.1 s, 0-12.7 s	Least significant bits of the Unix epoch timestamp of the GNSS fix, in tenths of a second, UTC
lat_acc	20	10	4.44 cm, ±23.3 km around reference	Latitude, RCE-encoded
lon_acc	20	30	4.44 or 8.9 cm, at least ±8.9 km	Longitude, RCE-encoded
sog	8	50	0.05 m/s, 0-47.6 m/s	Speed over ground. Unsigned exponential encoding, 6 mantissa bits (max 92.5 kn)
cog	9	58	0.703125°, 0-359.3°	Course over ground
hdg	9	67	0.703125°, 0-359.3°	True heading; only valid if the hdg_valid flag is set

Field	Width	Pos	Scaling, unit, range	Description
rot	7	76	0.5 °/s, ±116 °/s	Rate of turn, starboard positive. Signed exponential encoding, 4 mantissa bits
mode	2	83	enum, 0-3	Movement mode, see Field semantics
pos_acc	4	85	0.01 m, 0.01-327.67 m	95% horizontal position accuracy, unsigned exponential encoding without mantissa bits; 0 = unknown. See Field semantics
flags	3	89	flags	Bit 0: hdg_valid · Bit 1: rtk_fix · Bit 2: no_track
reserved	4	92	-	Set to 0

Heading and rate of turn let a rules engine reconstruct luffs, tacks and mark roundings between fixes. Nodes without a compass clear hdg_valid; receivers then use cog.

Information message (msg_type 1)

Library: mandatory (obl_frame; obl_geo for the antenna-to-hull reduction)

The information message links the radio address to identification and static boat data. One message is sent every 15 s, cycling through the parts in the order 0, 1, 0, 2, 0, 3. Part 0 thus repeats every 30 s and the other parts every 90 s; a new receiver identifies a boat within 30 s.

Field	Width	Pos	Description
ver	3	0	Protocol version, set to 1
part	2	3	0: sail number and class · 1: boat or skipper name · 2: MMSI · 3: dimensions

Field	Width	Pos	Description
payload (part 0)	91	5	class_id, 10 bits (registry-assigned class code); boat_type, 4 bits (0: unknown · 1: dinghy · 2: keelboat · 3: multihull · 4: board · 5: support or committee boat · 6: robotic mark · 7: land station · 8-15: reserved); sail number, 8 characters × 6 bits; coarse_lat, 6 bits, and coarse_lon, 6 bits, bits 18-23 of the scaled coordinates (see Relative coordinate encoding); perf_level, 3 bits, OBL performance tier of the device (1-3, 1 = highest; 0 = not declared); remaining bits reserved
payload (part 1)	91	5	name, 14 characters × 6 bits; remaining bits reserved
payload (part 2)	91	5	mmsi, 30 bits, the boat's AIS MMSI as a binary number, 0 if none assigned; remaining bits reserved
payload (part 3)	91	5	loa, 10 bits, 0.05 m units, length overall; beam, 8 bits, 0.05 m units; ant_bow, 10 bits, 0.05 m units, GNSS antenna distance aft of the bow; ant_height, 9 bits, 0.1 m units, antenna height above the waterline; remaining bits reserved

Characters use a 6-bit charset: 0 space, 1-26 A-Z, 27-36 0-9, 37 hyphen, 38 slash, 39 period; other codes are reserved. Boats with an MMSI send it in part 2, so that OBL and AIS data can be merged. Part 3 gives boat and antenna geometry, so that receivers can reduce the antenna position to bow and hull outline: OCS detection and umpiring judge the hull, not the antenna. An antenna more than 2 m above the water moves sideways with heel, by about 3.5 m at 7 m height and 30° heel; such devices shall compensate heel and pitch with their IMU and report the position of the hull below the antenna.

Telemetry message (msg_type 2)

Telemetry carries sensor data at up to 5 Hz. It shares slots with tracking, so its rate is limited by the event settings, and it is switched off under high load (see Capacity and scaling). It has no minimum rate.

Field	Width	Pos	Description
ver	3	0	Protocol version, set to 1
time	6	3	See Time under Field semantics
subtype	5	9	0: wind · 1: attitude · 2: rig loads · 3: water data (speed through water, depth) · 4: system status · 5-15: reserved · 16-31: vendor-specific
data	82	14	Subtype-specific

Subtype layouts. Subtypes 0 and 1 are examples; subtype 4 is normative. Remaining bits are reserved.

Subtype	Field	Width, coding
0: wind (example)	awa	9 bits, signed, 0.703125°, apparent wind angle
	aws	8 bits, 0.25 kn, apparent wind speed
	twd	9 bits, 0.703125°, true wind direction
	tws	8 bits, 0.25 kn, true wind speed
1: attitude (example)	heel	8 bits, signed, 0.5°
	trim	8 bits, signed, 0.5°
4: system status	battery	7 bits, 1 %
	thrust	7 bits, 1 % of maximum propulsion
	nav_state	3 bits: 0 idle · 1 transiting · 2 holding station · 3 propulsion stopped · 4 fault · 5-7 reserved
	dist_target	10 bits, 1 m, distance to the commanded target; 1023 = none
	flags	4 bits: bit 0 low battery · bit 1 off station · bit 2 GNSS degraded · bit 3 capsized

System status is the standard health report of unmanned nodes such as robotic marks. Off-station and capsized conditions should also raise the header urgency.

RTK correction message (msg_type 3)

Library: mandatory (obl_rtk, fragmentation and reassembly); satellite selection at the RTK base node: recommended

RTCM 3.x corrections from a venue RTK base node are split into 10-byte fragments and broadcast in the RTK slots (LR slots 4-11, see Timing), so that they reach the whole course, including boats with low antennas. Boats and marks get RTK positioning without a return channel. In Europe the RTK base node may transmit at up to 500 mW ERP. The stream is not subject to rate adaptation.

Field	Width	Pos	Description
ver	3	0	Protocol version, set to 1
stream_id	2	3	Correction stream identifier (multiple RTK base nodes per venue)
seq	7	5	Fragment sequence number, wraps around
len	4	12	Number of valid bytes in data (10 except in the last fragment)
data	80	16	RTCM 3.x byte stream fragment

Content

The correction source is a local base station or a network correction service (NTRIP, virtual reference station for the course area centre), relayed through a transmit-capable gateway node that acts as RTK base node and holds its certificate. The stream is filtered to the content below either way. The RTK base node sends one complete epoch every 2 s: MSM4 observations of two constellations, at most 8 satellites per constellation above 10° elevation, two frequencies each. The sender chooses the two constellations (GPS, Galileo or BeiDou) to suit the venue and its receivers. Both constellations of an epoch are sent together; alternating constellations between epochs is not permitted. GLONASS is not used: its FDMA signals need receiver-specific bias handling (RTCM message 1230).

RTCM message	Content	Rate
1074, 1094 or 1124	MSM4 observations of the two chosen constellations (GPS, Galileo, BeiDou)	every 2 s

RTCM message	Content	Rate
1005	Stationary antenna reference point	every 10 s
1033	Antenna and receiver descriptors	every 30 s

One epoch is about 280 bytes, so the stream averages about 140 bytes/s, or 14-16 LR packets per second including key disclosures. This is about 5.5 % duty cycle for the RTK base node, within the 10 % limit of the European LR band. Generic compression such as Deflate is not used: MSM data is already bit-packed, the gain is below 10 %, and a stateful decompressor does not survive lost fragments.

An RTK base node shall transmit only in the RTK slots and only while the event settings announce rtk = 1 (see Timing).

Framing

The sender validates the CRC-24 of each RTCM frame and removes the 3 header bytes (preamble 0xD3, reserved bits, 10-bit length) and the 3 CRC bytes. Only the RTCM message is fragmented, followed by a 3-byte MAC if the stream is authenticated (see Authentication). Each message is fragmented separately. A fragment with len < 10 closes the message; a message whose length is a multiple of 10 is closed by an additional fragment with len = 0.

The receiver reassembles the fragments in seq order, removes and verifies the MAC if present, adds preamble and reserved bits, sets the length field, recomputes the CRC-24 and passes the frame to its GNSS receiver. A gap in seq discards the current message; the receiver then skips fragments up to and including the next closing fragment. Fragment integrity is covered by the radio CRC and FEC.

Corrections arrive at most 2 s old. Short-baseline RTK normally keeps its fixed solution with corrections of this age. The accuracy achieved on moving boats shall be confirmed by trials before events rely on it for performance tier 1.

Race management message (msg_type 4)

Library: mandatory (obl_frame); application logic: recommended (obl_rc), mandatory for systems used as ARMS Category 2

The race committee sends race management messages (subtypes 0-7) from the committee node or a gateway node; boat nodes send their reports (subtypes 8-15) in the same message type. The committee messages carry the official race state: start

sequences, recalls, course changes, postponement and abandonment, mark assignments, commands to robotic marks and event settings; boat reports carry the start result of each boat. On-board countdowns and OCS detection are based on them. The messages are event-driven and refreshed at least every 10 s for late joiners.

Field	Width	Pos	Description
ver	3	0	Protocol version, set to 1
time	6	3	See Time under Field semantics
subtype	4	9	0: start sequence · 1: mark purposes · 2: course change · 3: postpone/abandon/shorten · 4: mark command · 5: event settings · 6: start result (individual recall) · 7: reserved · 8: start report, from boat nodes · 9: finish report, from boat nodes, layout reserved · 10-15: reserved
data	83	13	Subtype-specific

Subtype layouts. Remaining bits are reserved.

Subtype	Field	Width, coding
0: start sequence	race_id	8 bits
	state	3 bits: 0 idle · 1 warning · 2 preparatory · 3 one minute · 4 started · 5 general recall · 6 postponed · 7 abandoned
	prep_flag	3 bits: 0 none · 1 P · 2 I · 3 Z · 4 U · 5 black flag · 6-7 reserved
	start_time	17 bits, scheduled start, seconds of the UTC day
	course_id	6 bits
	class_id	10 bits, class starting in this race (as in the information message); 0 = all classes
1: mark purposes	course_id	6 bits

Subtype	Field	Width, coding
	count	2 bits, number of assignments in this packet (up to 2)
	per assignment	mark address, 24 bits; purpose, 6 bits: 0 start pin end · 1 start committee end · 2-33 course marks 1-32 · 34 finish 1 · 35 finish 2 · 36-63 reserved
2: course change	race_id	8 bits
	course_id	6 bits
	purpose	6 bits, mark whose position changes (codes as in subtype 1)
	bearing	9 bits, 0.703125°, true bearing from the preceding mark to the new position; 511 = not given
	length	2 bits: 0 unchanged or not given · 1 longer · 2 shorter · 3 reserved
3: postpone, abandon, shorten	race_id	8 bits; 255 = all races of the course area
	action	4 bits: 0 postpone · 1 postpone, further signals ashore · 2 postpone, no more racing today · 3 abandon · 4 abandon, further signals ashore · 5 abandon, no more racing today · 6 shorten course · 7 postponement removed · 8-15 reserved
	signal_time	17 bits, time of the signal, seconds of the UTC day
	purpose	6 bits, for action 6 the mark where the course is shortened (codes as in subtype 1); otherwise 0
4: mark command	address	24 bits, mark address
	cmd	2 bits: 0 navigate to target · 1 hold current position · 2 stop propulsion · 3 sound horn

Subtype	Field	Width, coding
	lat_acc, lon_acc	20 + 20 bits, target position, RCE-encoded
	speed	6 bits, 0.25 m/s, transit speed over ground
5: event settings	event_id	12 bits
	auth_mode	2 bits: 0-2 (see Authentication); 3 reserved
	enc	1 bit, 1 = boat nodes encrypt tracking and telemetry (see Encryption); requires auth_mode 1 or 2
	dev_threshold	6 bits, 0.1 m, EBR deviation threshold; 0 = default of the device's performance tier
	hdg_threshold	4 bits, 5°, EBR heading threshold; 0 = default (15°), 15 = disabled
	rate_max	3 bits, highest periodic rate: 0 0.125 Hz · 1 0.25 Hz · 2 0.5 Hz · 3 1 Hz · 4 2 Hz · 5-7 reserved
	telemetry_max	4 bits, highest telemetry rate in Hz (at most 5); 0 = disabled
	relay	1 bit, request relay = 1 for tracking messages
	rtk	1 bit, RTK stream active: LR slots 4-11 are reserved for RTK
	hb_set	2 bits, HB channel set of the course area (see Channel plan); 0 = default set
	area_lat, area_lon	20 + 20 bits, centre of the course area, encoded as in the tracking message
	area_radius	6 bits, 100 m, radius of the course area; 0 = not given
6: individual recall	race_id	8 bits
	count	2 bits, number of boats listed in this packet (0-3)

Subtype	Field	Width, coding
	address × count	24 bits each, addresses of boats identified on the course side
	pending	1 bit, 1 = start reports missing, result under review
8: start report	race_id	8 bits
	result	2 bits: 0 clear · 1 over · 2 too close to call · 3 no result (no fix or no line)
	margin	14 bits, signed, 0.01 m, foremost hull point to the line at the starting signal; course side positive
	acc	4 bits, pos_acc code of the boat at the signal
	line_acc	4 bits, worse pos_acc of the two line ends
	flags	3 bits: bit 0 rtk_fix · bit 1 hdg_valid · bit 2 interpolated

- Subtype 0: each boat computes its start result from the start sequence and the start line positions (see subtype 8). Boats show their OCS status only from the committee's start result (subtype 6); their own margin may be shown as information. General recall is state 5.
- Subtype 1: several packets cover the full course. A mark may have several purposes, for example start pin end and finish 2, and then receives one assignment per purpose.
- Subtype 2: the new mark position follows from the tracking messages of the mark, or from a subtype 1 update if a different mark is used.
- Subtype 4: target and speed apply to cmd 0 only. A mark confirms execution through its tracking and telemetry messages.
- Subtype 5: sent at least every 30 s, in LR slots as well, so that devices learn their HB channel set on the common channels. Devices apply the settings of the course area whose geofence contains their position, as long as they receive the sender, and return to their defaults 10 minutes after losing it.

- Subtype 6 implements ARMS OCS identification as the start result. A boat is identified when, at her starting signal, any part of her hull is on the course side of the starting line; the hull position follows from the antenna position and the geometry in information message part 3. The committee builds the list from the start reports (subtype 8). For a registered boat of the starting class without a valid report by cycle S+2, the committee application computes the margin at the signal from the tracking messages it received, with the uncertainty pos_acc + deviation threshold + SOG × time to the nearest fix; if the margin lies clearly outside this bound, the result is decided from tracking. The remaining boats are missing: the result is broadcast with pending = 1, and the race officer reviews them on a list with their last positions and the track from 10 s before to 10 s after the signal, relative to the line. The review log records whether each result came from a report, from tracking or from the race officer. The committee sends the first start result in its infrastructure slots of cycle S+2 (HB F0 and LR F2), 2.0-2.75 s after the signal, so that all boats receive it at the same moment, and repeats it at up to 1 Hz as long as boats remain identified or pending, at most until 4 minutes after the starting signal. Boats that have returned and started correctly are removed from the list. More than three boats are spread over several packets. A message with count = 0 and pending = 0 is the all-clear. Receivers shall show the boat's own status to its crew.
- Subtype 8 is the boat's start report. The device computes the position of the foremost hull point at the starting signal, interpolated from its recorded fixes just before and after the signal, and the line from the latest tracking messages of the two line ends, interpolated to the signal. The result is over if margin > acc + line_acc, clear if margin < −(acc + line_acc), and too close to call otherwise; the race officer decides the last group. The report is sent in an HB contention slot of cycle S+1 with a fresh draw and repeated in F0 of cycle S+2; in these two cycles boats send no deviation reports. If two different reports arrive for one address, both are discarded and the boat counts as missing. From the preparatory signal until 1 minute after the start, the committee node and the start-line mark node send tracking at 2 Hz in infrastructure slots, with devices of the tier required of the fleet. Subtype 9, the finish report, follows the same pattern; its layout is reserved for the next revision.

Safety message (msg_type 5)

Safety messages implement check-in and check-out (presence control before, during and after racing) and distress acknowledgements. They are event-driven and may be repeated at up to 1 Hz to ensure delivery.

Field	Width	Pos	Description
ver	3	0	Protocol version, set to 1
time	6	3	See Time under Field semantics
subtype	4	9	0: check-in · 1: check-out · 2: distress acknowledgement · 3: assistance request · 4-15: reserved
event_id	12	13	Event/venue identifier from the event settings (race management message, subtype 5)
data	71	25	Subtype-specific; for subtype 2: address of the acknowledged station, 24 bits

Distress is not a separate message. A node in distress sets urgency = 3 and mode = 3 in its tracking messages, so every alert carries a current position. In distress, the node sends tracking at the highest periodic rate of its mode (one packet in each frame of its mode, 2 Hz), always with relay = 1, and ignores rate adaptation and EBR savings. For loss of GNSS time in distress see Timing.

Weather message (msg_type 6)

Race marks, committee boats and land stations broadcast local weather every 10 to 30 s. A sender clears the valid flag of each sensor it does not have. Marks send their position in tracking messages and battery and drift state in telemetry (subtype 4); mark purposes come from race management messages (subtype 1).

Field	Width	Pos	Scaling, unit	Description
ver	3	0	-	Protocol version, set to 1
time	6	3	0.1 s	See Time under Field semantics
wind_dir	7	9	2.8125°	True wind direction, relative to true north
wind_avg	7	16	0.5 kn	Average wind speed (1 min)
wind_gust	7	23	0.5 kn	Gust (1 min maximum)
air_temp	8	30	0.5 °C	Air temperature, signed

Field	Width	Pos	Scaling, unit	Description
water_temp	8	38	0.5 °C	Water temperature, signed
pressure	10	46	0.1 hPa	Barometric pressure, offset 950 hPa; 0 = unknown
flags	4	56	flags	Bit 0: wind valid · Bit 1: air_temp valid · Bit 2: water_temp valid · Bit 3: pressure valid
reserved	36	60	-	Set to 0

Authentication message (msg_type 7)

Authentication messages carry MACs, disclosed keys and key chain commitments (see Authentication). Only authenticating devices send them.

Field	Width	Pos	Description
ver	3	0	Protocol version, set to 1
subtype	2	3	0: MAC and key disclosure · 1: chain commitment · 2: key disclosure only · 3: reserved for extension
data	91	5	Subtype-specific

Subtype layouts. Remaining bits are reserved.

Subtype	Field	Width, coding
0: MAC and key disclosure	ref_type	3 bits, msg_type of the authenticated packet
	ref_time	7 bits, time field of that packet (6-bit time fields zero-extended; part number for information messages)
	mac	24 bits, MAC of that packet under the key of the current interval
	key	56 bits, disclosed chain key of the previous interval
1: chain commitment	part	5 bits

Subtype	Field	Width, coding
	data	86 bits, part of the commitment or certificate chain
2: key disclosure only	key	56 bits, disclosed chain key of the previous interval

Subtype 1 carries the commitment in parts 0-6: anchor interval index (32 bits), anchor key (56 bits) and an ECDSA P-256 signature (512 bits, r ‖ s) over sender address, anchor interval index and anchor key, 600 bits in total, the last part zero-padded. Devices send their certificate chain (own device certificate and issuer certificate) in parts 7-31, at least every 5 minutes; infrastructure nodes of an event with encryption also send the organizer's encryption key certificate; the certificate format is still to be defined. Subtype 2 is used by senders whose MAC travels with the data, such as RTK streams.

Field semantics

Time

The time field of the tracking message holds the 7 least significant bits of the Unix time in tenths of a second (UTC) of the GNSS fix, not of the transmission. Receivers with valid time resolve the 12.8 s ambiguity. At 25 m/s (49 kn) the quantization error of at most 0.05 s corresponds to 1.25 m. All other message types carry the 6 least significant bits of the same value (6.4 s ambiguity).

Movement mode

mode	Description
0	Undefined / unknown
1	Stationary (moored, ashore, mark boat on station)
2	Under way
3	Distress (capsized, man overboard, disabled)

Urgency

urgency	Description
0	Routine

urgency	Description
1	Elevated, race-critical phase (e.g. starting sequence); receivers should prioritize processing
2	Urgent, assistance requested; always sent with relay = 1
3	Distress, receivers shall alert their operators; relays shall always forward

Position accuracy (pos_acc)

pos_acc is the 95 % horizontal position accuracy, coded as unsigned exponential encoding without mantissa bits at a scale of 0.01 m: code c represents $(2^c - 1) \times 0.01$ m, and code 0 means unknown. Senders shall send the smallest code that is not below their accuracy estimate. The code values are 0.01, 0.03, 0.07, 0.15, 0.31, 0.63, 1.27, 2.55, 5.11, 10.23, 20.47, 40.95, 81.91, 163.83 and 327.67 m.

Flags

- hdg_valid: hdg holds a measured true heading, from a three-axis magnetometer with three-axis IMU or from dual-antenna GNSS. With a magnetometer, the gyro carries the heading over the short term and the magnetometer corrects it; the magnetometer shall be calibrated and tilt-compensated, and the magnetic heading corrected to true north with the declination from the World Magnetic Model, evaluated from the device's position and date. The heading is valid only while the magnetometer heading and the gyro-propagated heading agree within the provisional limit of the device's performance tier (see OBL performance tiers). The limit follows from its effect: a heading error e shifts the computed bow by ant_bow × sin(e); at 2 m antenna-to-bow distance, 2° is 7 cm and 5° is 17 cm, within each tier's position accuracy.
- rtk_fix: the position is an RTK fixed solution.
- no_track: the sender asks not to be stored or forwarded to public tracking outside the sporting context. Reception for race management and safety remains allowed.

Appendix B: Example integration into an existing device

This example adds OBL to a product that already has a GNSS receiver and a microcontroller, such as a race computer, an instrument display or a robotic race mark. A pre-certified radio module runs the complete OBL stack. The host exchanges messages

with it over a serial port and needs no RF design. The host provides:

- a UART (115.2 kbaud) for messages between host and module
- the 1PPS output of its GNSS receiver, wired to a module GPIO for slot timing
- NMEA or binary position data at 1-10 Hz over the UART
- a 3.3 V supply, 60 mA peak during transmission
- about 2 cm² of board space and an antenna position as described in Antenna and installation

The example uses the Ebyte E70-900T14S2, a CC1310 module for 861-922.5 MHz with CE and FCC modular certification. The CC1310 has an Arm Cortex-M3 application core and a separate radio core. The OBL reference firmware runs on the module and implements PHY and MAC: 2-GFSK at 300 kbit/s, slot timing from the 1PPS input, LDPC coding, whitening and the message set. Its 14 dBm output corresponds to the European 25 mW ERP limit. The host sends position and telemetry and receives race management, safety and RTK messages; RTK-capable hosts forward the corrections to their GNSS receiver. Indicative added bill of materials at 100 pieces:

Item	Example part	Price (USD)
OBL radio module	Ebyte E70-900T14S2 (TI CC1310, 868/915 MHz, 14 dBm, CE/FCC)	4.00
Antenna	868/915 MHz quarter-wave whip (RP-SMA), fixed flexible PCB antenna, or antenna integrated into the device, e.g. a centre-fed dipole	1.50
RF cabling	u.FL to SMA pigtail with bulkhead connector	1.00
RF protection	ESD diode and matching passives at the antenna port	0.30
Supply and glue	decoupling, host connector; LDO only if no 3.3 V rail exists	0.70
Total		approx. 7.50

With a modular FCC grant and a CE declaration, the product does not repeat radio type approval; the host needs a “Contains FCC ID” label, host emissions testing and a reference to the module declaration.

Appendix C: Reference implementations

This appendix gives reference implementations of the main algorithms in C99 and Python 3. They are informative (see Code fragments): compact, without error handling, written for clarity. For products, the official C libraries implement these algorithms (see Conformance disclosure). The C code assumes two's-complement arithmetic with arithmetic right shift. The Python code relies on floored division and non-negative modulo, which match the definitions directly.

Exponential encoding

Decoding uses the closed form from the Exponential encoding section. Encoding selects the smallest exponent that can represent the value and rounds to the nearest code point. The signed variants use the most significant bit as sign bit.

C:

```
uint32_t exp_decode(uint32_t code, size_t N, size_t N_m) {
  uint32_t e = code >> N_m;
  uint32_t m = code & ((1u << N_m) - 1);
  return (((1u << N_m) + m) << e) - (1u << N_m);
}

uint32_t exp_encode(uint32_t value, size_t N, size_t N_m) {
  uint32_t base = 1u << N_m;
  for (uint32_t e = 0; e < (1u << (N - N_m)); e++) {
    uint32_t m = ((value + base + ((1u << e) >> 1)) >> e) - base;
    if (m < base) return (e << N_m) | m;
  }
  return (1u << N) - 1;  /* clamp to the maximum code point */
}

uint32_t exp_encode_signed(int32_t value, size_t N, size_t N_m) {
  uint32_t s = value < 0 ? 1u << (N - 1) : 0;
  uint32_t mag = (uint32_t)(value < 0 ? -value : value);
  return s | exp_encode(mag, N - 1, N_m);
}

int32_t exp_decode_signed(uint32_t code, size_t N, size_t N_m) {
  int32_t v = (int32_t)exp_decode(code & ((1u << (N - 1)) - 1), N - 1, N_m);
  return (code >> (N - 1)) ? -v : v;
}
```

Python:

```
def exp_decode(code, n, n_m):
    e, m = code >> n_m, code & ((1 << n_m) - 1)
    return (((1 << n_m) + m) << e) - (1 << n_m)

def exp_encode(value, n, n_m):
    base = 1 << n_m
    for e in range(1 << (n - n_m)):
        m = ((value + base + (1 << e >> 1)) >> e) - base
        if m < base:
            return (e << n_m) | m
    return (1 << n) - 1  # clamp to the maximum code point

def exp_encode_signed(value, n, n_m):
    s = 1 << (n - 1) if value < 0 else 0
    return s | exp_encode(abs(value), n - 1, n_m)

def exp_decode_signed(code, n, n_m):
    v = exp_decode(code & ((1 << (n - 1)) - 1), n - 1, n_m)
    return -v if code >> (n - 1) else v
```

Relative coordinate encoding

With D = 4 or 8 and m = 2^{20}, the floored division is an arithmetic shift in C and the modulo a bit mask. Python's // and % match the definitions. Raw coordinates are in units of 1E-7 degrees.

C:

```
#define RCE_M (1 << 20)                    /* grid modulus m */

unsigned rce_lon_shift(int32_t lat_raw) {  /* 2: D=4, 3: D=8 */
  return (lat_raw < 0 ? -lat_raw : lat_raw) >= 660000000 ? 3 : 2;
}

uint32_t rce_encode(int32_t c_raw, unsigned shift) {
  int32_t q = (c_raw + (1 << (shift - 1))) >> shift;  /* floored div by D */
  return (uint32_t)q & (RCE_M - 1);                   /* mathematical mod */
}

int32_t rce_decode(uint32_t c_comp, int32_t ref_raw, unsigned shift) {
  int32_t q_ref = (ref_raw + (1 << (shift - 1))) >> shift;
  int32_t delta = (int32_t)((c_comp - (uint32_t)q_ref + RCE_M / 2)
                            & (RCE_M - 1)) - RCE_M / 2;
  return (q_ref + delta) * (1 << shift);
}
```

Python:

```
RCE_M = 1 << 20                        # grid modulus m

def rce_lon_divider(lat_raw):
    return 8 if abs(lat_raw) >= 660_000_000 else 4

def rce_encode(c_raw, d):
    return ((c_raw + d // 2) // d) % RCE_M

def rce_decode(c_comp, ref_raw, d):
    q_ref = (ref_raw + d // 2) // d
    delta = (c_comp - q_ref + RCE_M // 2) % RCE_M - RCE_M // 2
    return (q_ref + delta) * d
```

Extrapolation-based reporting

The prediction advances the last decoded position along its course over ground at its speed over ground. Transmitter and receivers shall use the decoded (quantized) values. With IEEE 754 double precision, platform differences stay orders of magnitude below the deviation threshold.

C:

```
/* lat, lon: 1e-7 deg; sog: m/s; cog: deg; dt: s since the fix */
void ebr_predict(int32_t lat, int32_t lon, double sog, double cog,
                 double dt, int32_t *lat_p, int32_t *lon_p) {
  const double RAD = 0.017453292519943295;
  const double UNITS_PER_M = 1e7 / 111194.9266;  /* 1e-7 deg per metre */
  double d = sog * dt;
  *lat_p = lat + (int32_t)lround(d * cos(cog * RAD) * UNITS_PER_M);
  *lon_p = lon + (int32_t)lround(d * sin(cog * RAD) * UNITS_PER_M
                                 / cos(lat * 1e-7 * RAD));
}
```

Python:

```
from math import cos, sin, radians

UNITS_PER_M = 1e7 / 111194.9266    # 1e-7 deg units per metre

def ebr_predict(lat, lon, sog, cog, dt):
    d = sog * dt                   # metres travelled since the fix
    lat_p = lat + round(d * cos(radians(cog)) * UNITS_PER_M)
    lon_p = lon + round(d * sin(radians(cog)) * UNITS_PER_M
                        / cos(radians(lat * 1e-7)))
    return lat_p, lon_p
```

CRC

CRC-16/CCITT, polynomial 0x1021, initial value 0xFFFF, over sync word, header and payload in transmission order. The C function processes one byte, the Python function a whole buffer.

C:

```
uint16_t crc_obl(uint16_t crc, uint8_t data) {
  crc ^= (uint16_t)data << 8;
  for (int i = 0; i < 8; i++)
    crc = (crc & 0x8000) ? (crc << 1) ^ 0x1021 : (crc << 1);
  return crc;  /* init with 0xFFFF */
}
```

Python:

```
def crc_obl(data: bytes) -> int:
    crc = 0xFFFF
    for byte in data:
        crc ^= byte << 8
        for _ in range(8):
            crc = ((crc << 1) ^ 0x1021 if crc & 0x8000 else crc << 1) &
0xFFFF
    return crc
```

PN9 data whitening

PN9 generator ($x^9 + x^5 + 1$, seed 0x1FF) as clocked by common sub-GHz transceivers. The output bytes are XORed with header, payload, CRC and FEC parity.

C:

```
uint8_t pn9_next(uint16_t *state) {  /* init *state with 0x1FF */
  uint8_t out = *state & 0xFF;
  for (int i = 0; i < 8; i++) {
    uint16_t fb = ((*state & 1) ^ ((*state >> 5) & 1)) << 8;
    *state = (*state >> 1) | fb;
  }
  return out;  /* XOR with the next data byte */
}
```

Python:

```
def pn9_stream(n: int) -> bytes:
    state, out = 0x1FF, bytearray()
    for _ in range(n):
        out.append(state & 0xFF)
        for _ in range(8):
            fb = ((state & 1) ^ (state >> 5 & 1)) << 8
            state = (state >> 1) | fb
    return bytes(out)  # XOR byte-wise with the data
```

Appendix D: Sailing instructions guide for ARMS

Events that use OBL as ARMS need sailing instructions that change some racing rules. This informative guide gives text that organizers adapt to their event, in the style of the World Sailing Sailing Instructions Guide. Rule numbers refer to the RRS 2025-2028.

Class rules. Where class rules restrict equipment, they must require or permit the ARMS device, for example through a class rule change applied through the notice of race under RRS 87. Such changes can also exclude the device from the class weight rules.

Recognition. Class rules and sailing instructions refer to a World Sailing recognised ARMS, and World Sailing may withhold approval of class rule changes that require a non-recognised ARMS. Policy D12 defines Category 1 (decision support: race officials keep the final decision) and Category 2 (rule-determining: outputs may decide designated race management or officiating questions, through RRS changes in the notice of race or sailing instructions). World Sailing's draft 2026 categories add Category 0 for testing only. Each OBL-based system needs its own recognition; this specification defines the common protocol. ARMS.3 and ARMS.4 make the ARMS determinative and therefore need a Category 2 system: ARMS.3 (start timing) and ARMS.4 (OCS identification) with devices of OBL performance tier 2 or 1. With a Category 1 system, the race committee uses the OBL identification as information and keeps the decision, and ARMS.4 is omitted.

Sailing instruction text:

- ARMS.1 [NP][DP] Each boat shall carry an ARMS device using OBL, installed [position, e.g. on the aft side of the mast] with the mounting system specified by the organizing authority. The organizing authority may provide devices.
- ARMS.2 [NP][DP] Boats shall not move, modify, disable, remove or interfere with the ARMS device except as instructed by the organizing authority, and shall promptly notify the race committee of any device issue.
- ARMS.3 The official time of the start is the time provided by the ARMS. Visual signals are displayed for information only [or: will not be displayed; sound signals will be made]. This changes RRS 26.
- ARMS.4 A boat may be identified by the ARMS when, at her starting signal, any part of her hull is on the course side of the starting line. The race committee broadcasts identified boats as individual recall within 3 seconds of the starting signal. The indication of a boat's own device is advisory; the race committee's identification is decisive. Flag X will not be displayed and no sound signal will be made. This changes RRS 29.1 [and 30.1-30.4 if used for these].
- ARMS.5 If the race committee displays flag W with a long sound signal before the warning signal of a race, ARMS.3 and ARMS.4 do not apply to that race.
- ARMS.6 Alleged failure, inaccuracy or malfunction of the ARMS device or system is not grounds for requesting redress. This changes RRS 61.1(a).

- ARMS.7 For the purpose of RRS 41, information broadcast by OBL in clear and displayed by ARMS devices is information freely available to all boats. [If tracking is encrypted:] Tracking data of other boats is not freely available until the organizing authority publishes the keys.
- ARMS.8 If the ARMS has no start report from a boat, the race committee may review her start from tracking data. If it finds that a boat on the course side of the starting line at her starting signal was not notified, it may signal a general recall until the first boat has rounded the first mark. This changes RRS 29.2. After that, the race committee may abandon the race under RRS 32.1. [Events whose rules do not permit a late general recall delete the second and third sentences.]

Notes. Sailing instructions for other ARMS often also require a daily check-in of the device with the race committee, a signal to restart devices, and notices of required firmware updates. With OBL the race committee receives every device directly, so the check-in can be automatic (safety message, subtype 0), and no restart is needed (see Zero configuration). Firmware updates that affect the event should still be announced on the official notice board. Events with authentication enabled use verified data only for OCS identification, finishing and scoring (see Authentication).

Appendix E: World Sailing Policy D12 compliance

This appendix maps the principles (3.2) and possible requirements (3.3) of World Sailing Policy D12 of 27 February 2026 to this specification. It is informative. Requirements defined later by the World Sailing Executive Office will be added.

D12	Requirement	OBL
3.2(a)	Alignment with the RRS	Sailing instructions guide (Appendix D); OCS identification by hull position; race committee decides with Category 1 systems. Covered.
3.2(b)	Fairness and equal treatment	Race management and safety data are broadcast in clear to all boats (RRS 41); boat data is clear for all boats or encrypted for all and published at a time the organizer sets (Encryption); same protocol for all vendors. Covered.

D12	Requirement	OBL
3.2(c)	Transparency and auditability	Open specification, Implementation Conformance Statement, gateway records of the last known state, on-board record of every fix at tiers 1 and 2, publication of encryption keys at a time the organizer sets. Open: logging of race committee decisions and ARMS determinations.
3.2(d)	Data integrity, security and reliability	Authentication with trust chain (Authentication); CRC and FEC. Covered.
3.2(e)	Operational resilience, offline capability	Direct boat-to-boat and boat-to-committee links; fixed degradation order (Capacity and scaling). Covered.
3.3(a)	Positional and timing accuracy	OBL performance tiers, RTK, start report with margin and accuracy, time base ±5 µs. Open: position and heading accuracy measurements.
3.3(b)	Secure and encrypted communications	Authentication of all critical data (Authentication); optional encryption of boat tracking and telemetry, with keys published after racing (Encryption). Covered.
3.3(c)	Resilience to interference, data loss, degradation	FEC, listen-before-talk, extrapolation-based reporting, degradation order. Open: capacity simulation.
3.3(d)	Controlled updates, version traceability	ICS names specification and firmware version. Open: versioning and firmware update rules.
3.3(e)	Consistent, repeatable determinations	Prediction from decoded values; mandatory library modules used unchanged in every product; reference implementations (Appendix C). Open: test vectors and conformance tests.

D12	Requirement	OBL
3.3(f)	Interfaces for race committee, umpires, sailors	Race management messages including the start result and the review list for missing reports; user interfaces are product-specific. Partly covered.
3.3(g)	Global regulatory compliance	Channel plan per regulatory region. Open: regulatory review.
3.3(h)	Operation without internet	No internet needed on the water; gateways are optional. Covered.
3.3(i)	Interfaces for media and broadcast	Gateways with UDP/MQTT, CBOR and JSON (Gateways). Covered.
4.1(g)	Risk of cost inflation	Open protocol, multi-source hardware, USD 20-30 electronics. Covered.